EE108
Optoelectronics Intellisense (OI)

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Artificial Intelligence (AI)
Data / Algorithms / Computing

Computing / Algorithms / Data

1. Preprocessing input data
2. Training the learning model
3. Storing the trained learning model
4. Deployment of the model

CPU

CPU (Central Processing Unit) is the primary hardware of the computer that executes the instruction for computer programs. All the basic arithmetic, logic, controlling, and the CPU handles input/output functions of the program.

CPU runs the operating system, continually receiving inputs and providing output to the users. A CPU contains at least one processor. Some high-end companies also build CPUs with eight processors.

Popular Manufacturers: Intel, AMD, Qualcomm, NVIDIA, IBM, Samsung, Hewlett-Packard, VIA, etc

AI-enabled chips

We need high-performance CPUs for **computer vision**, **natural language processing (NLP)**, or **speech recognition**.

AI-enabled chips become a solution to this challenge. These chips make CPUs "intelligent" for optimizing their tasks. As a result, CPUs can work for their duties individually and improve their efficiency. New AI technologies will require these chips to solve complicated tasks and perform them faster.

Companies like Facebook, Amazon, and Google are increasing their investments in AI-enabled chips. These chips will assist next-generation databases for faster query processing and predictive analytics. Industries like healthcare and automobile heavily rely on these chips for delivering intelligence.

Advances in Graphics Processing Units (GPU)

GPU are one of the most commercially used type of AI enabled chips.

Rendering an image requires simple computing power but needs to be done on a large scale very quickly. GPUs are the best option for such cases because they can process thousands of simple tasks simultaneously. As new technologies in GPU renders better-quality images since they do these simple tasks a lot faster.

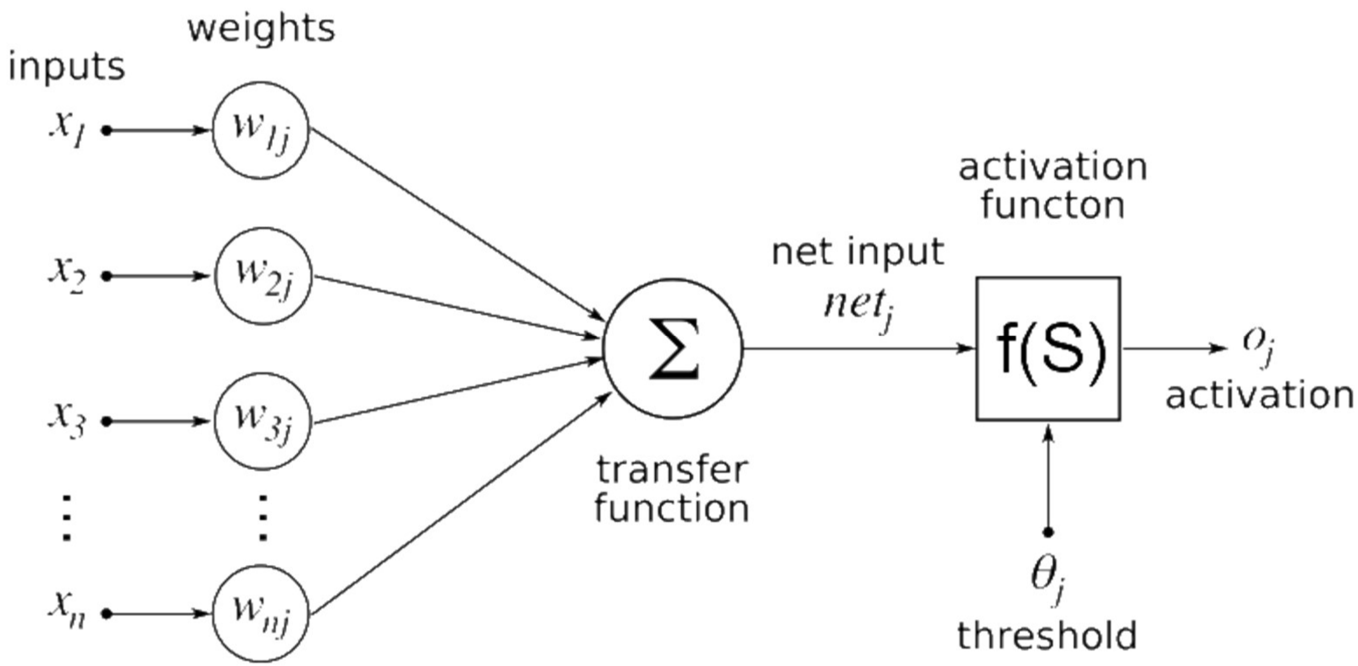
Modern GPUs have become powerful enough to be used for tasks beyond image rendering, such as cryptocurrency mining or machine learning. While CPUs usually used to do these tasks, data scientists discovered that these are repetitive parallel tasks. Thus, GPUs are widely used in AI models for efficient learning.

Popular GPU Manufacturers: NVIDIA, AMD, Broadcom Limited

Advances in Graphics Processing Units (GPU)

When you train a deep learning model, two main operations are performed: Forward Pass & Backward Pass

In forward pass, input is passed through the neural network and after processing the input, an output is generated. Whereas in backward pass, we update the weights of neural network on the basis of error we get in forward pass.



Advances in Graphics Processing Units (GPU)

Both of these operations are essentially matrix multiplications. A simple matrix multiplication can be represented by this →

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & \\ & \end{bmatrix}$$

$$1 \cdot 7 + 2 \cdot 9 + 3 \cdot 11 = 58$$

In a neural network, we can consider first array as input to the neural network, and the second array can be considered as weights of the network.

This seems to be a simple task. Now just to give you a sense of what kind of scale deep learning – VGG16 (a convolutional neural network of 16 hidden layers which is frequently used in deep learning applications) has ~140 million parameters; aka weights and biases. Now think of all the matrix multiplications you would have to do to pass just one input to this network! It would take years to train this kind of systems if we take traditional approaches.

Advances in Graphics Processing Units (GPU)

Researchers at Stanford built the same system in terms of computation to train their deep nets using GPU. They reduced the costs to just $33K ! This system was built using GPUs, and it gave the same processing power as Google's system.

	Google	Stanford
Number of cores	1K CPUs = 16K cores	3GPUs = 18K cores
Cost	$5B	$33K
Training time	week	week

Note: A processor core is an individual processor within a CPU. Many computers today have multi-core processors (e.g. INTEL i9 CPU has 8 cores). In the past, computer CPUs only had a single core.

Advances in Graphics Processing Units (GPU)

GPU vs CPU

GPU	CPU
• hundreds of simpler cores	• few very complex cores
• thousand of concurrent hardware threads	• single-thread performance optimization
• maximize floating-point throughput	• transistor space dedicated to complex ILP
• most die surface for integer and fp units	• few die surface for integer and fp units

GPU is mostly used for gaming and doing complex simulations. These tasks and mainly graphics computations, and so GPU is graphics processing unit. If GPU is used for non-graphical processing, they are termed as GPGPUs – general purpose graphics processing unit

ILP-Instruction-level parallelism
fp-floating point

TPU

Tensor Processing Unit (TPU) is an application-specific integrated circuit, to accelerate the AI calculations and algorithm. An open-source machine learning platform, with state of the art tools, libraries, and community, so the user can quickly build and deploy ML apps.

Cloud TPU allows you to run your machine learning projects on TPU using TensorFlow. The models who used to take weeks to train on GPU or any other hardware can put out in hours with TPU.

Manufacturer: Only Google makes them since 2015.

CPU vs GPU vs TPU

CPU	GPU	TPU
Several core	Thousands of Cores	Matrix based workload
Low latency	High data throughput	High latency
Serial processing	Massive parallel computing	High data throughput
Limited simultaneous operations	Limited multitasking	Suited for large batch sizes
Large memory capacity	Low memory	Complex neural network models

Edge Computing

This is a distributed computing framework that brings enterprise applications closer to data sources (e.g. IoT devices or local edge servers). An effective edge computing model should address network security risks, management complexities, and the limitations of latency and bandwidth.

Gartner estimates that by 2025, 75% of data will be processed outside the traditional data center or cloud.

A viable model should consider the followings:

- Manage your workloads across all clouds and on any number of devices
- Deploy applications to all edge locations reliably and seamlessly
- Maintain openness and flexibility to adopt to evolving needs
- Operate more securely and with confidence

Fog Computing

CLOUD LAYER

Big Data Processing
Business Logic
Data Warehousing

FOG LAYER

Local Network
Data Analysis & Reduction
Control Response
Virtualization/Standardization

EDGE LAYER

Large volume Real-time Data Processing
At Source/On Premises Data Visualization
Industrial PCs
Embedded Systems
Gateways
Micro Data Storage

Fog computing will process, filter, manage, and analyze data gathered by various sensors and IoT devices. Therefore, fog computing can enable intelligent applications to run at the edge in real-time by bringing powerful computing at the edge.

Fog computing offloads the computation task from the cloud down to the local area network (LAN). However, by implementing an additional layer between the cloud and the edge, fog computing is adding complexity to the IoT network architecture.

Quantum computing

Traditional computer systems work with binary states; 0 and 1. However, quantum computing takes this to another level and works with quantum mechanics. This enables quantum systems to work with **qubits**, instead of bits. While bits consist of 0 and 1, qubits consist of 0, 1 or an additional state, which includes both at the same time. This additional state enables quantum computing to be open to new possibilities and provide faster computation for certain tasks. These tasks include neural network optimizations and digital approximations.

IBM states that it will be possible to build a quantum computer with 50-100 qubits in the next 10 years. When we consider that the 50-qubit quantum computer works faster than today's best 500 supercomputers, there is significant potential for quantum computing to provide additional computing power.

Ref: IBM predictions on quantum computing

Quantum Computer @ China

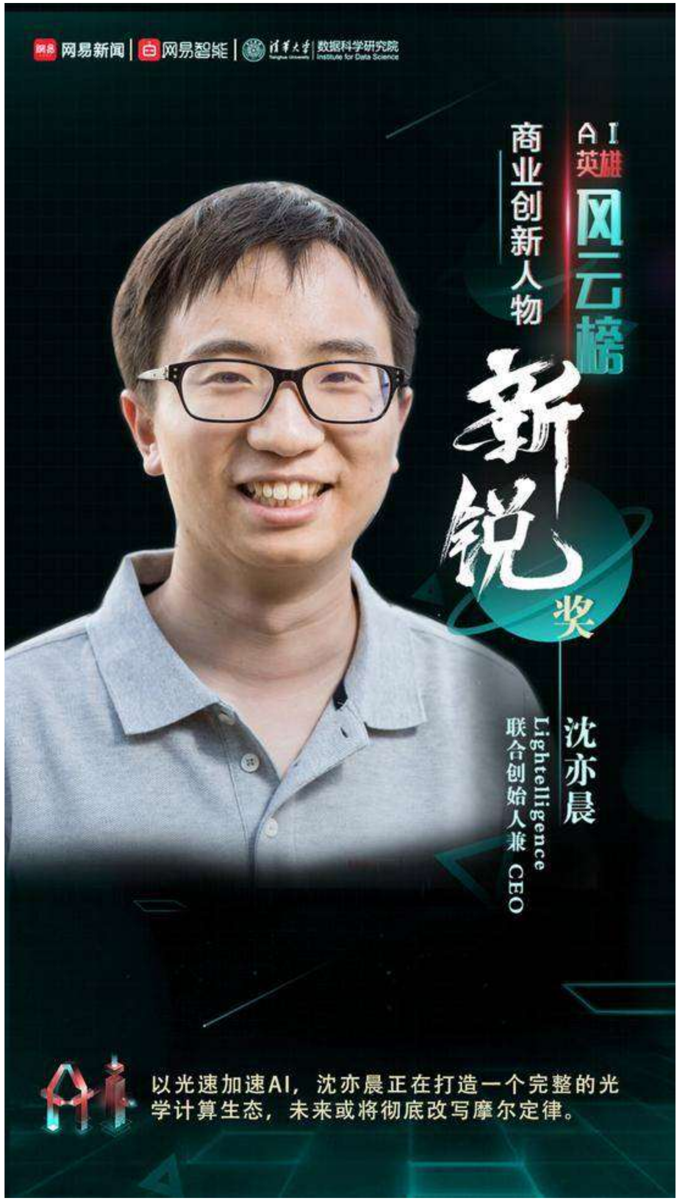
On 7 Dec 2020 a team of Chinese scientists has developed the most powerful quantum computer in the world, capable of performing at least one task 100 trillion times faster than the world's fastest supercomputers.

In 2019, Google said it had built the first machine to achieve "quantum supremacy," the first to outperform the world's best supercomputers at quantum calculation. The Chinese computer makes its calculations (about the behavior of light particles) using optical circuits. The Chinese team reported their quantum computer, named Jiuzhang (九章), is 10 billion times faster than Google's. A description of Jiuzhang and its feat of calculation was published 3 Dec 2020 in the journal Science. Assuming both claims hold up, Jiuzhang would be the second quantum computer to achieve quantum supremacy anywhere in the world.

Ref: https://www.livescience.com/china-quantum-supremacy.html

Quantum Computer

- Quantum computers can solve problems that are impossible or would take a traditional computer an impractical amount of time (a billion years) to solve.
- Change the landscape of data security by developing virtually unbreakable encryption. Predictions are that we would create hack-proof replacements.
- Every day, we produce 2.5 exabytes of data. That number is equivalent to the content on 5 million laptops. Quantum computers will make it possible to process the amount of data we're generating.
- Superposition is the term used to describe the quantum state where particles can exist in multiple states at the same time, and which allows quantum computers to look at many different variables at the same time.
- Reduce power consumption anywhere from 100 up to 1000 times because quantum computers use quantum tunnelling.
- IBM's computer Deep Blue defeated chess champion in 1997. It was able to gain a competitive advantage because it examined 200 million possible moves each second. A quantum machine would be able to calculate 1 trillion moves per second!



2018年12月28日，由网易智能、清华大学数据科学研究院和24家评审机构共同评出的「2018中国AI英雄风云榜」年度人物榜单揭晓，10位人工智能领域的从业者获奖。其中，Lightelligence 联合创始人兼CEO沈亦晨凭借光子芯片开拓性研究获得了本次评选的商业新锐奖。

什么是光子芯片和光子计算？在2016年，沈亦晨所在的MIT团队提出用光子代替电子为理论基础的计算芯片架构，沈亦晨曾通俗的解释说，光子芯片的运算原理有点像眼镜镜片，镜片会对穿过它的光波有一个复杂的运算，而光子芯片使用多光束干涉技术，其干涉条纹信息反映了所需计算的结果。他们把这种架构芯片称之为“可编程纳米光子处理器”。

https://www.lightelligence.ai/team

Advances in Algorithms

Explainable AI (XAI)

One of the main weak points of AI models is its **complexity**. Building and understanding an AI model requires a certain level of programming skills and, it costs time to digest the workflow of the model. As a result, companies usually benefit from the results of AI models without understanding their workflow.

To solve this challenge, Explainable AI makes these models understandable by anyone. XAI has three main goals:

- How the AI model affects developers and users
- How it affects data sources and results
- How inputs lead output

As an example, AI models will be able to diagnose diseases in the future. However, doctors also need to know how AI comes up with the diagnosis. With XAI, they can understand how AI makes its analysis and explain the situation to their patients accordingly.

Transfer learning

Transfer learning is a machine learning method that enables users to benefit from a previously used AI model for a different task. In several cases, it is clever to use this technique for the following reasons:

- Some AI models aren't easy to train and can take weeks to work properly. When another task comes up, developers can choose to adopt this trained model, instead of creating a new one. This will save time for model training.
- There might not be enough data in some cases. Instead of working with a small amount of data, companies can use previously trained models for more accurate results.

As an example, an AI model that is well-trained to recognize different cars can also be used for trucks. Instead of starting from scratch, the insight gained from cars will be beneficial for trucks.

Reinforcement learning (RL)

Reinforcement learning is a subset of machine learning which aims AI agent to take action for maximizing its reward. Rather than traditional learning, RL doesn’t look for patterns to make predictions. It makes sequential decisions to maximize its reward and it learns by experience.

Today, the most common example of RL is Google’s DeepMind AlphaGo which has defeated the world’s number one Go (圍棋) player in two consecutive games. In the future, RL may also be available in fully automated factories and self-driving cars.

Self-Supervised Learning (Self-Supervision)

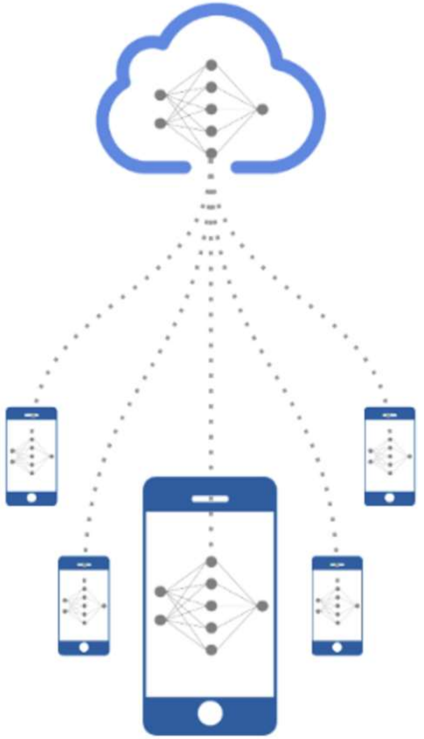
Self-supervised learning (or self-supervision) is a form of autonomous supervised learning. Unlike supervised learning, this technique doesn't require humans to label data, and it handles the labeling task by itself. According to Yann LeCun, Facebook VP and chief AI scientist, self-supervised learning will play a critical role in understanding human-level intelligence.

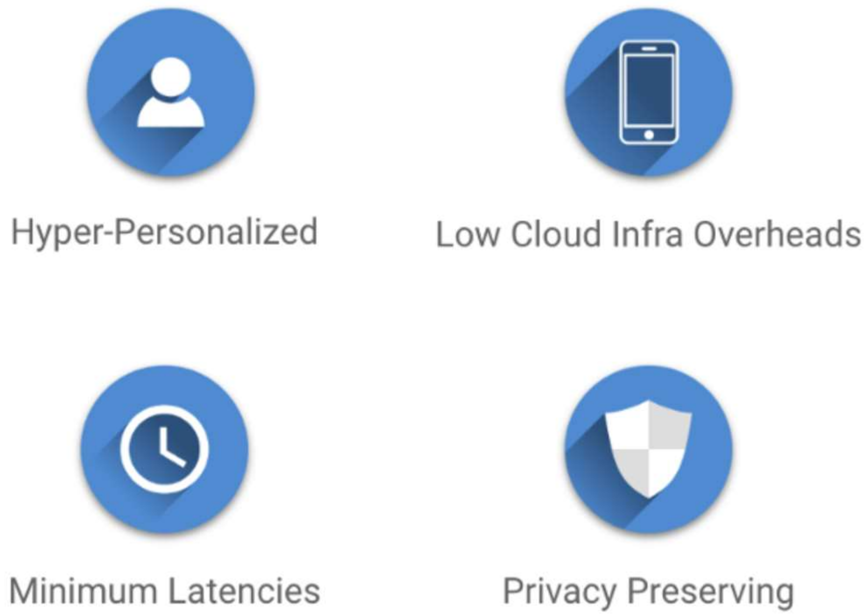
While this method is mostly used in computer vision and NLP tasks like image colorization or language translation today, it is expected to be used more widely in our daily lives. Some future use cases of self-supervised learning include:

- **Healthcare:** This technique can be used in robotic surgeries and estimating the dense depth in monocular endoscopy.
- **Autonomous driving:** It can determine the roughness of the terrain in off-roading and depth completion while driving.

Federated Learning

Federated learning enables machine learning models obtain experience from different data sets located in different sites (e.g. local data centers, a central server) without sharing training data. This allows personal data to remain in local sites, reducing possibility of personal data breaches.





Neural network compatibility and integration

Choosing the best neural network framework is a challenge for data scientists. As there are many AI tools in the market, it is important to choose the best AI tool for implementing the neural network framework. However, once a model is trained in one AI tool, it is hard to integrate the model into other frameworks.

To solve this problem, Facebook, Microsoft, and Amazon are cooperating to build Open Neural Network Exchange (ONNX) to integrate trained neural network models across multiple frameworks. In the future, ONNX is expected to become an essential technology for the industry.

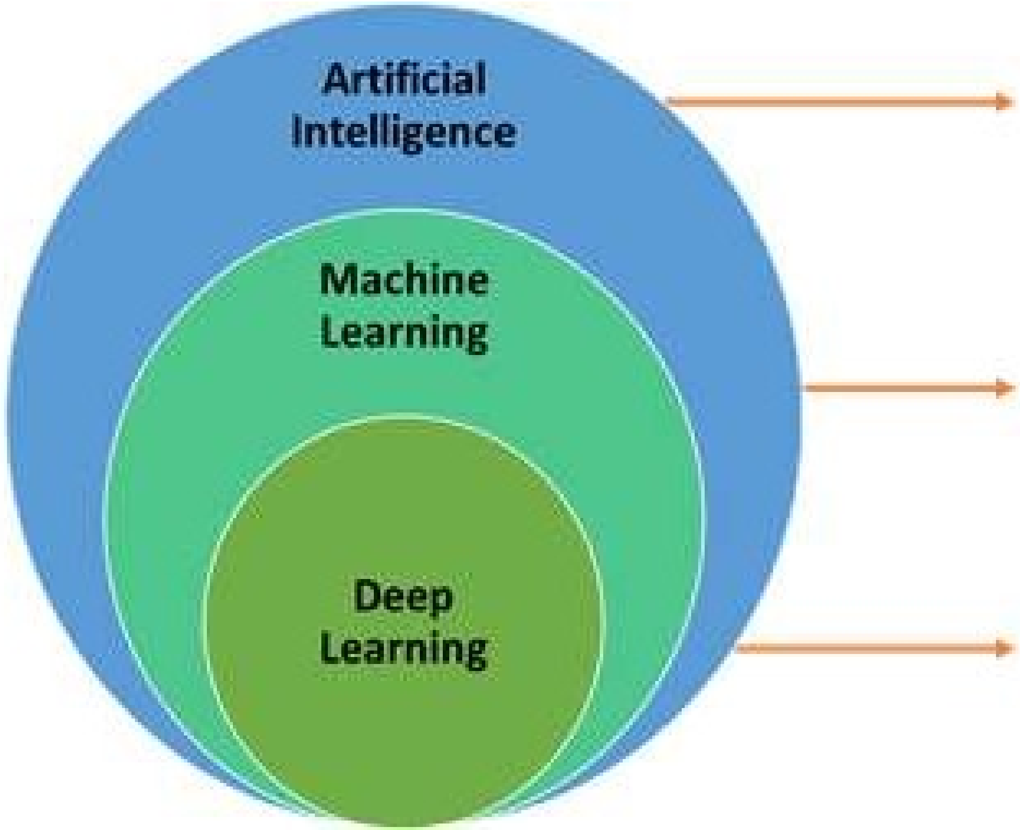
Ref: Artificial Intelligence (AI): In-depth Guide


DATA


2020 every MINUTE of the DAY
PRESENTED BY DOMO
ZOOM HOSTS 208,333 PARTICIPANTS IN MEETINGS
REDDIT SEES 479,452 PEOPLE ENGAGE WITH CONTENT
NETFLIX USERS STREAM 404,444 HOURS OF VIDEO
DOORDASH DINERS ORDER 555 MEALS
INSTAGRAM USERS POST 347,222 STORIES
YOUTUBE USERS UPLOAD 500 HRS OF VIDEO
TWITTER GAINS 319 NEW USERS
CONSUMERS SPEND $1,000,000 ONLINE
MICROSOFT TEAMS CONNECT 52,083 USERS
FACEBOOK USERS SHARE 150,000 MESSAGES
LINKEDIN USERS APPLY FOR 69,444 JOBS
TIKTOK IS INSTALLED 2,704 TIMES
VENMO USERS SEND $239,196 WORTH OF PAYMENTS
SPOTIFY ADDS 28 TRACKS TO ITS MUSIC LIBRARY
INSTAGRAM BUSINESS PROFILE ADS SEE 138,889 CLICKS
AMAZON SHIPS 6,659 PACKAGES
1,388,889 PEOPLE MAKE VIDEO/ VOICE CALLS
$3,805 IS SPENT ON MOBILE APPS
WHATSAPP USERS SHARE 41,666,667 MESSAGES
FACEBOOK USERS UPLOAD 147,000 PHOTOS

Machine Learning

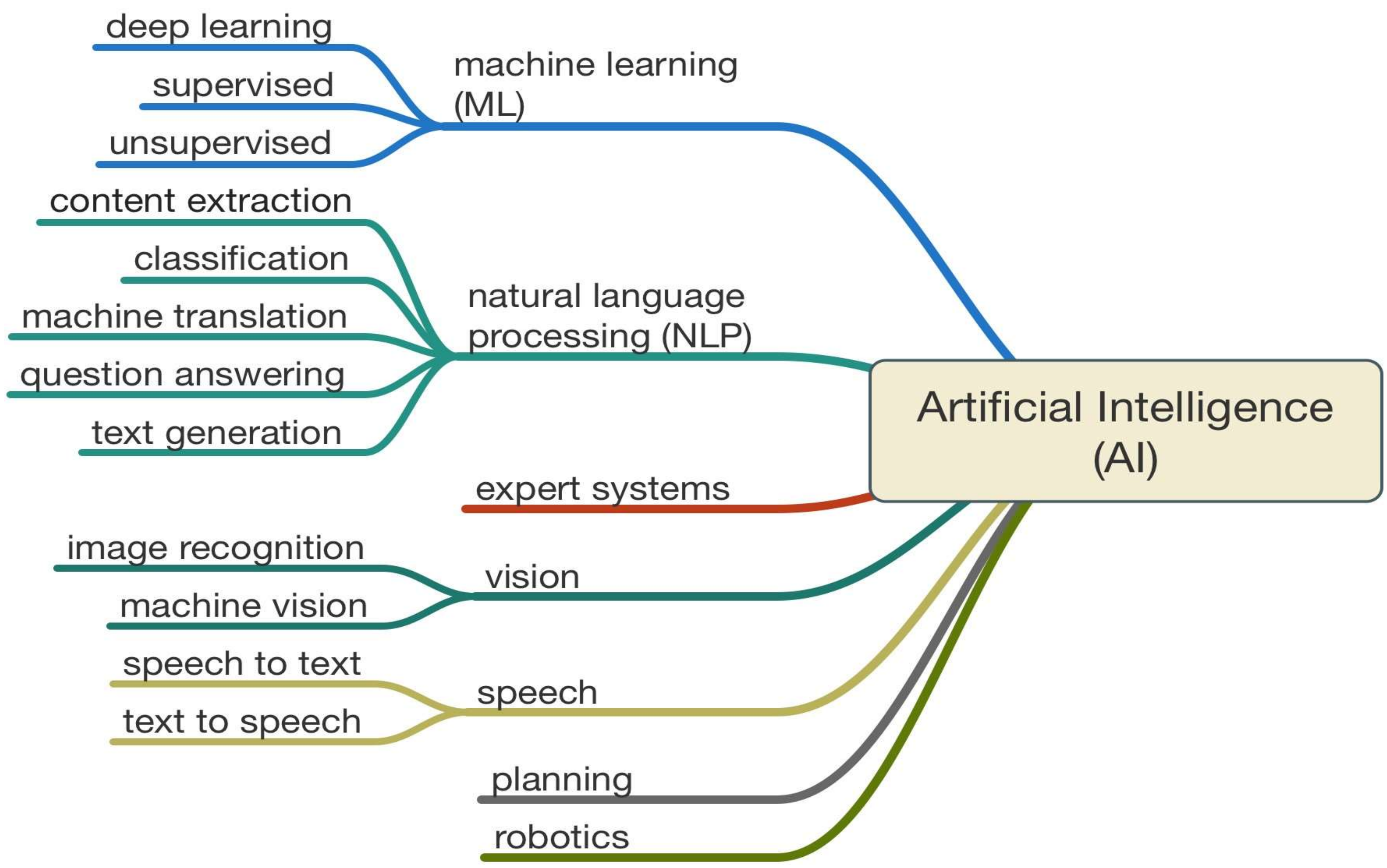
What is Machine Learning ?

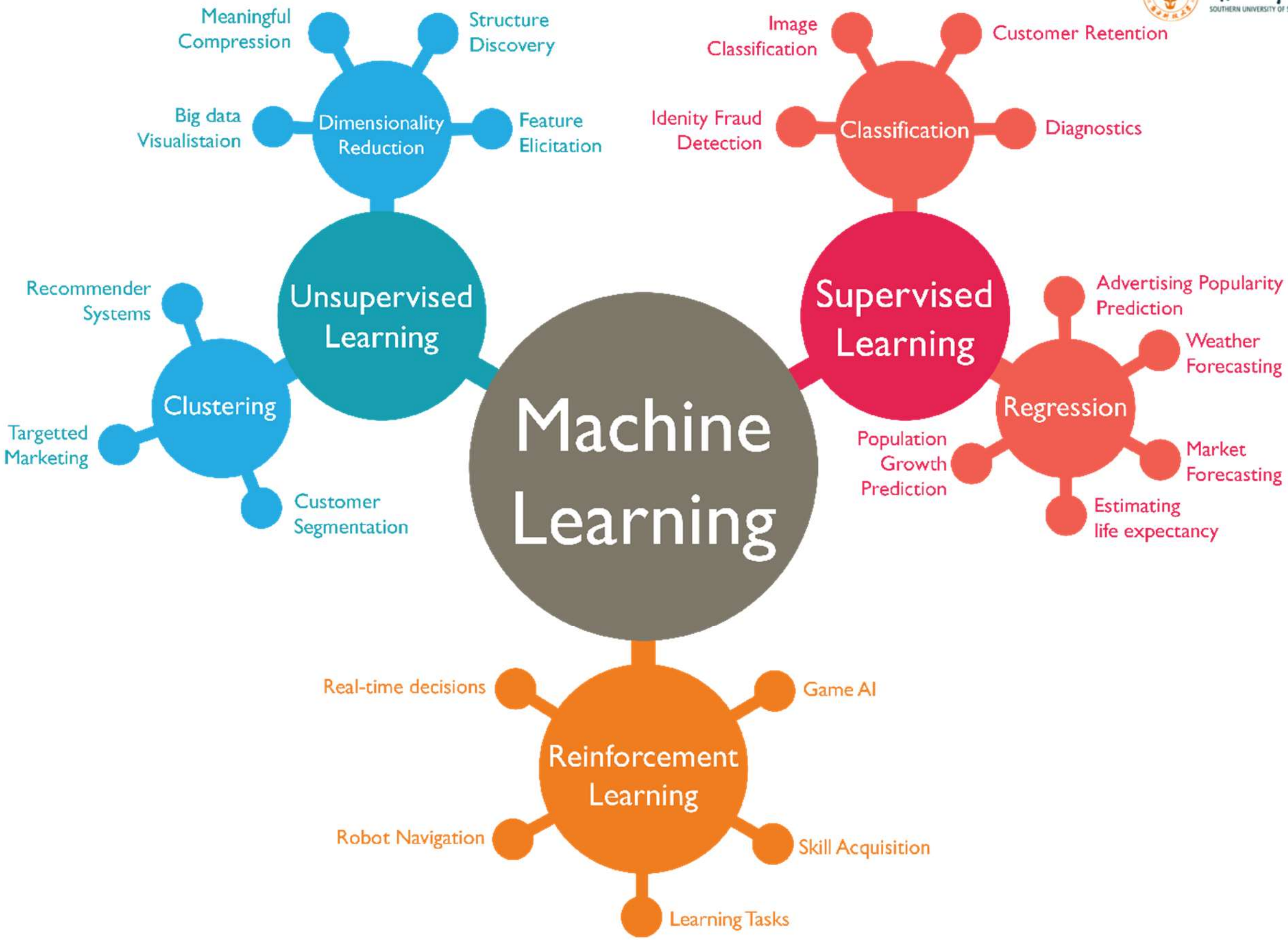


AI is a technique that enables computers and machines to mimic human intelligence, using logic, if-then rules, decision trees, and machine learning.

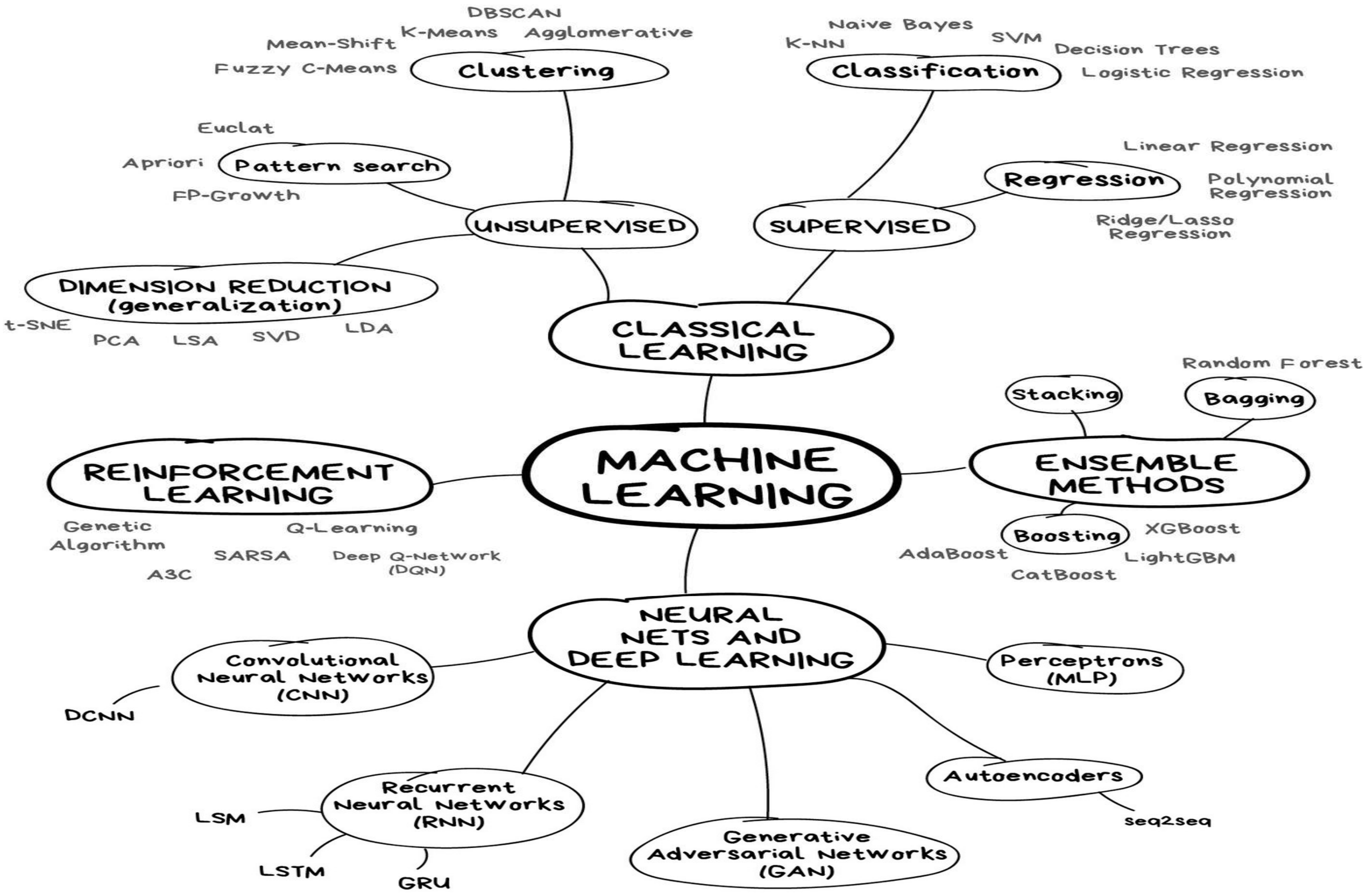
A subset of AI that includes abstruse statistical techniques that enables machines to improve at tasks with experience.

The subset of machine learning composed of algorithms that permits software to train itself to perform tasks, like speech and image recognition, by exposing multi-layered neural networks to vast amounts of data


deep learning
supervised
unsupervised
machine learning (ML)
content extraction
classification
machine translation
question answering
text generation
natural language processing (NLP)
expert systems
image recognition
machine vision
vision
speech to text
text to speech
speech
planning
robotics
Artificial Intelligence (AI)

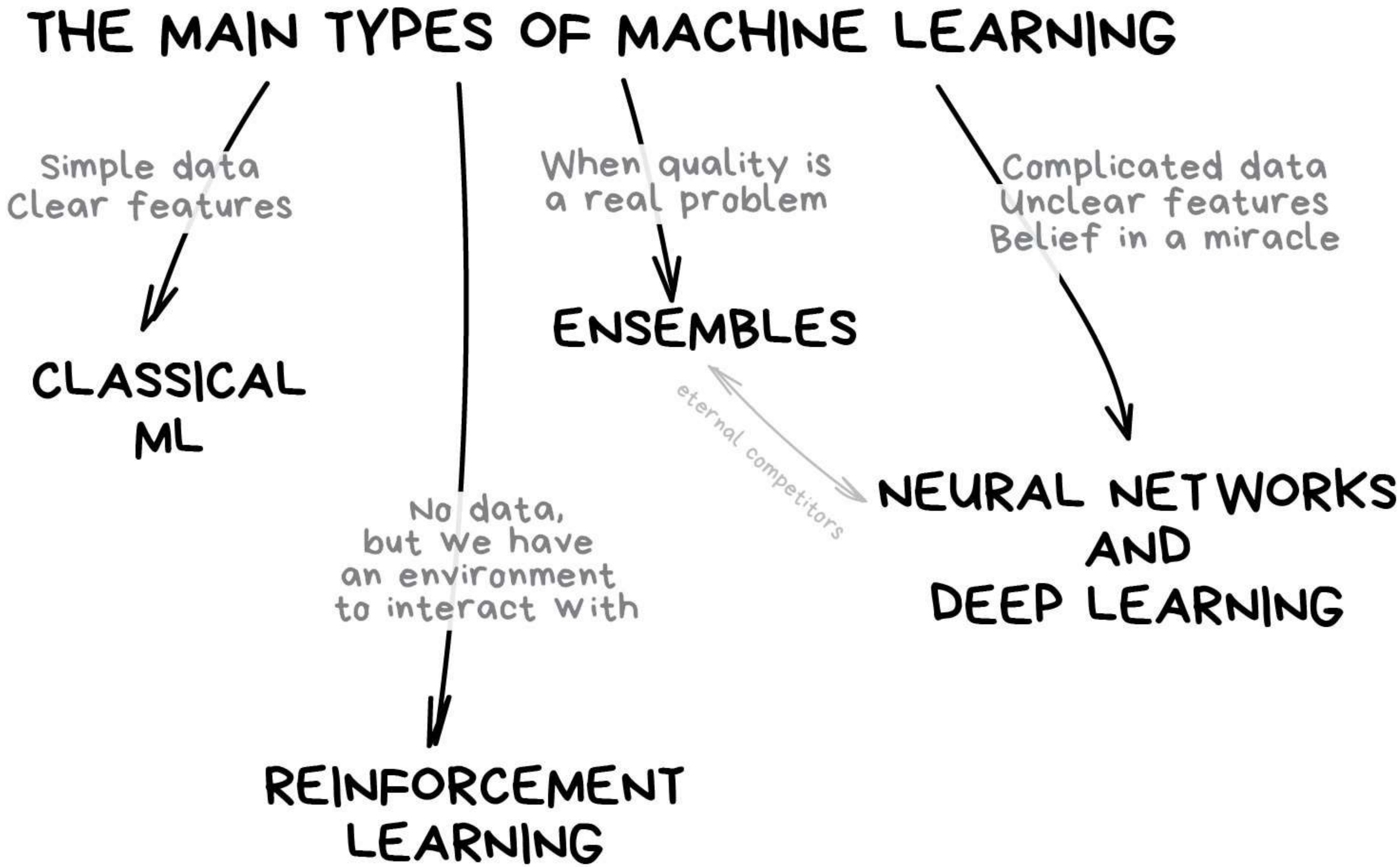

Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

Different Algorithm for different types of ML

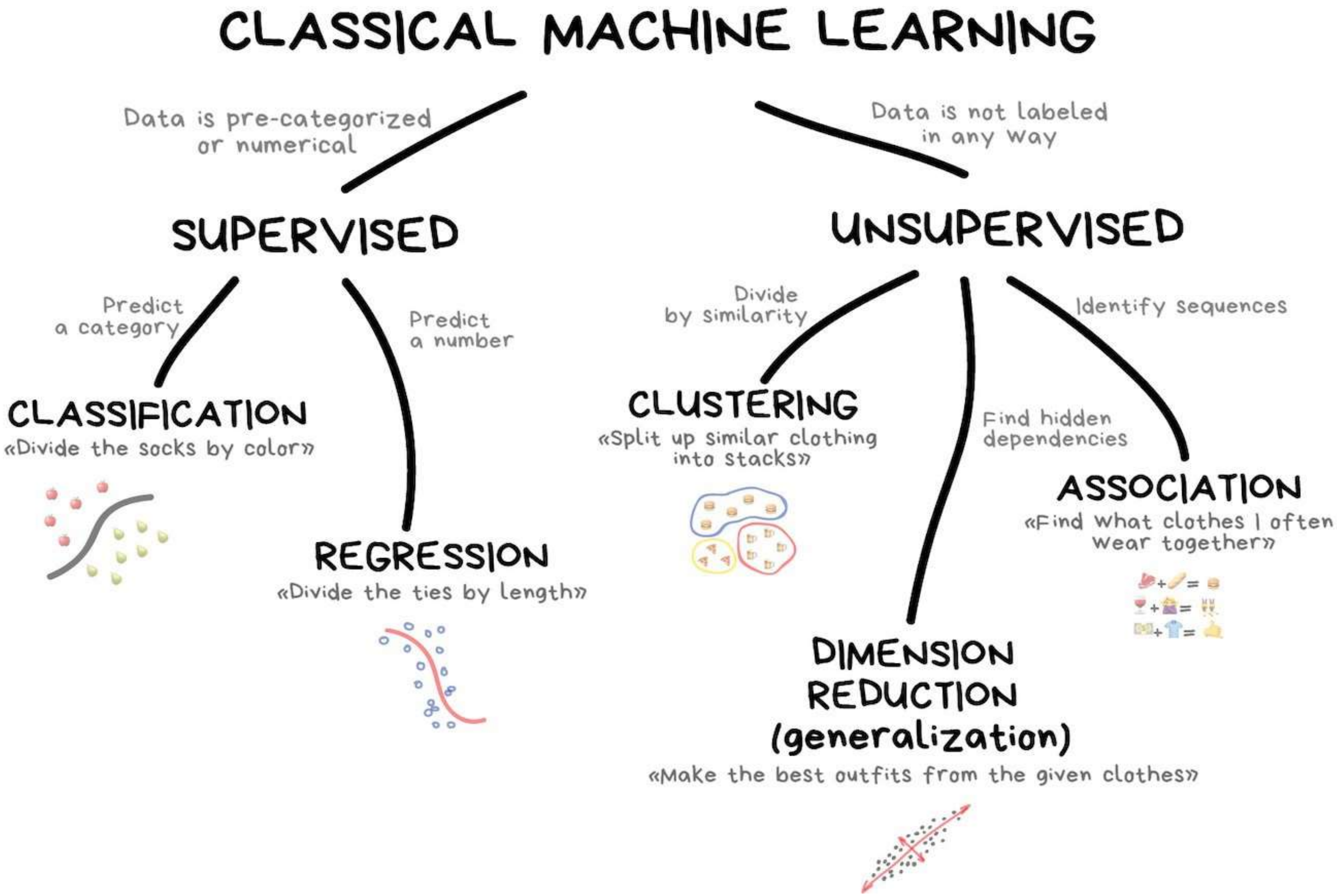


What type of ML to be used?

南方科技大学 SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY | 电子与电气工程系 DEPARTMENT OF ELECTRONIC AND ELECTRICAL ENGINEERING

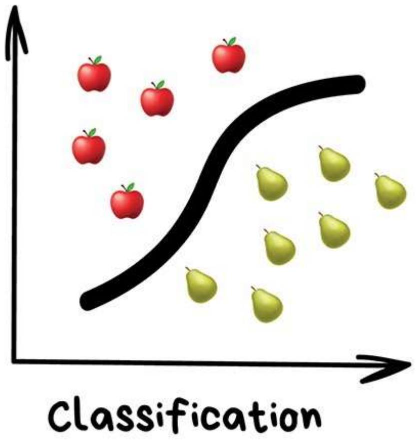


Classical ML



Supervised-Classification

"Splits objects based at one of the attributes known beforehand. Separate socks by based on color, documents based on language, music by genre"



Today used for:

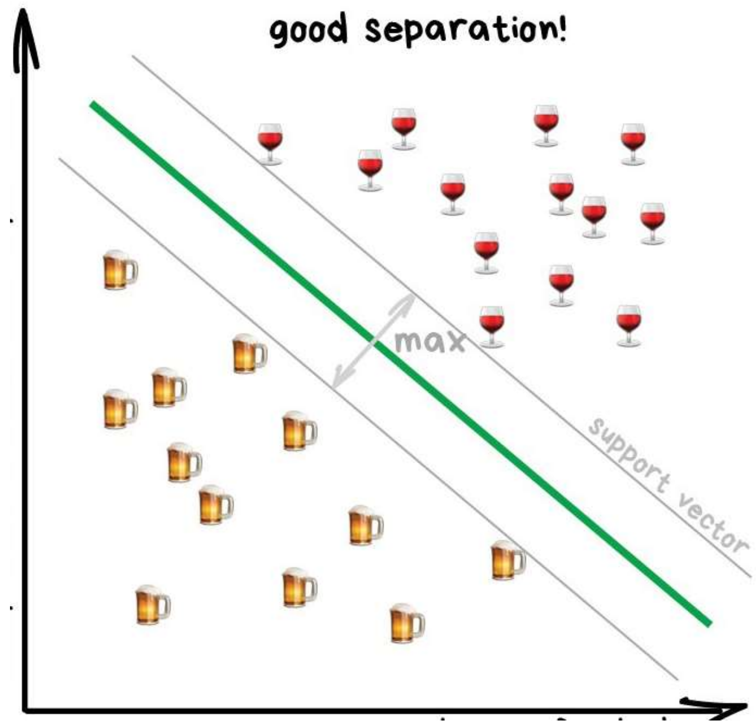
- Spam filtering
- Language detection
- A search of similar documents
- Recognition of handwritten characters and numbers

Popular algorithms: Naive Bayes, Decision Tree, Logistic Regression, K-Nearest Neighbours, Support Vector Machine

Supervised-Classification: Support-Vector Machine

Support-Vector Machines (SVM) is rightfully the most popular method of classical classification

- It's trying to draw two lines between your data points with the largest margin between them.
- Texts, numbers, tables – classical approach
    - ✓ Smaller model
    - ✓ Learn faster
    - ✓ Work on clearly
- Pictures, videos, complicated big data - NN

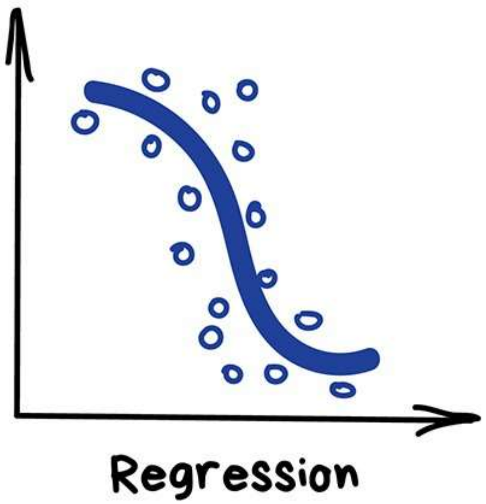


Supervised-Regression

Classification where we forecast a number instead of category.

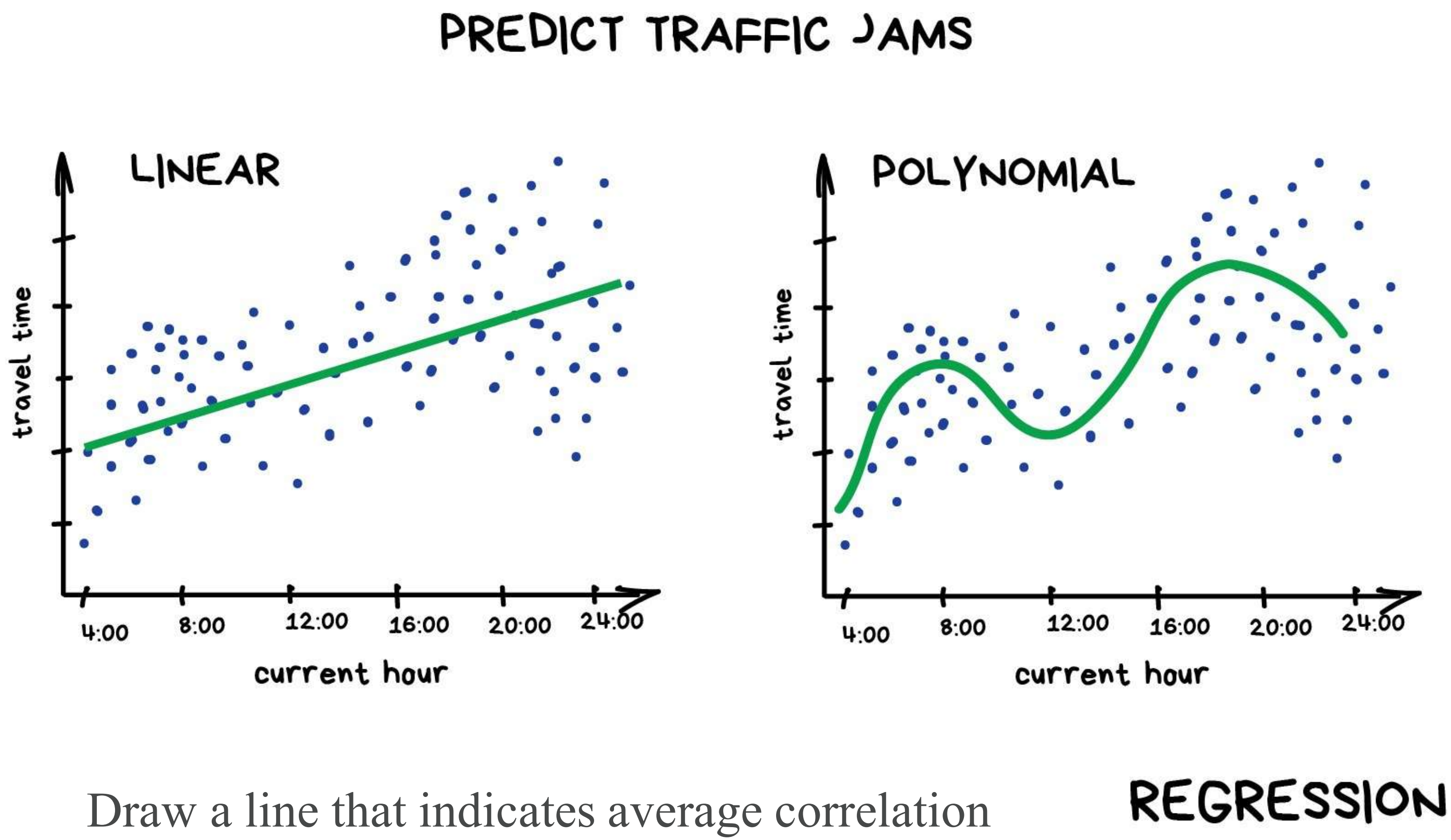
Today this is used for:

- Stock price forecasts
- Demand and sales volume analysis
- Medical diagnosis
- Any number-time correlations



Popular algorithms are Linear and Polynomial regressions

Supervised-Regression



Draw a line that indicates average correlation

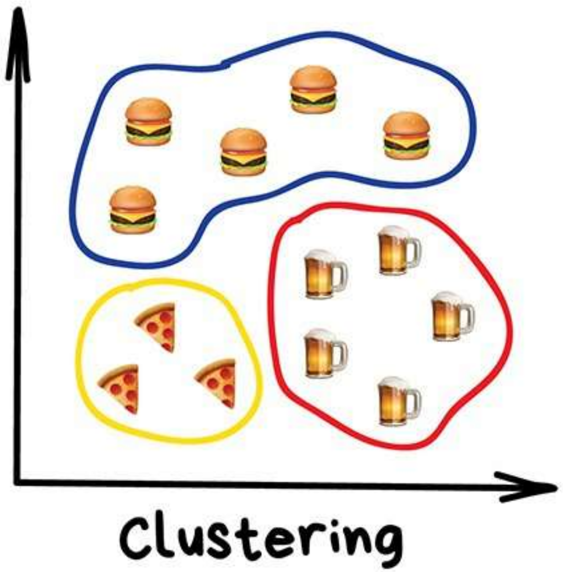
Unsupervised-Clustering

"Divides objects based on unknown features. Machine chooses the best way"

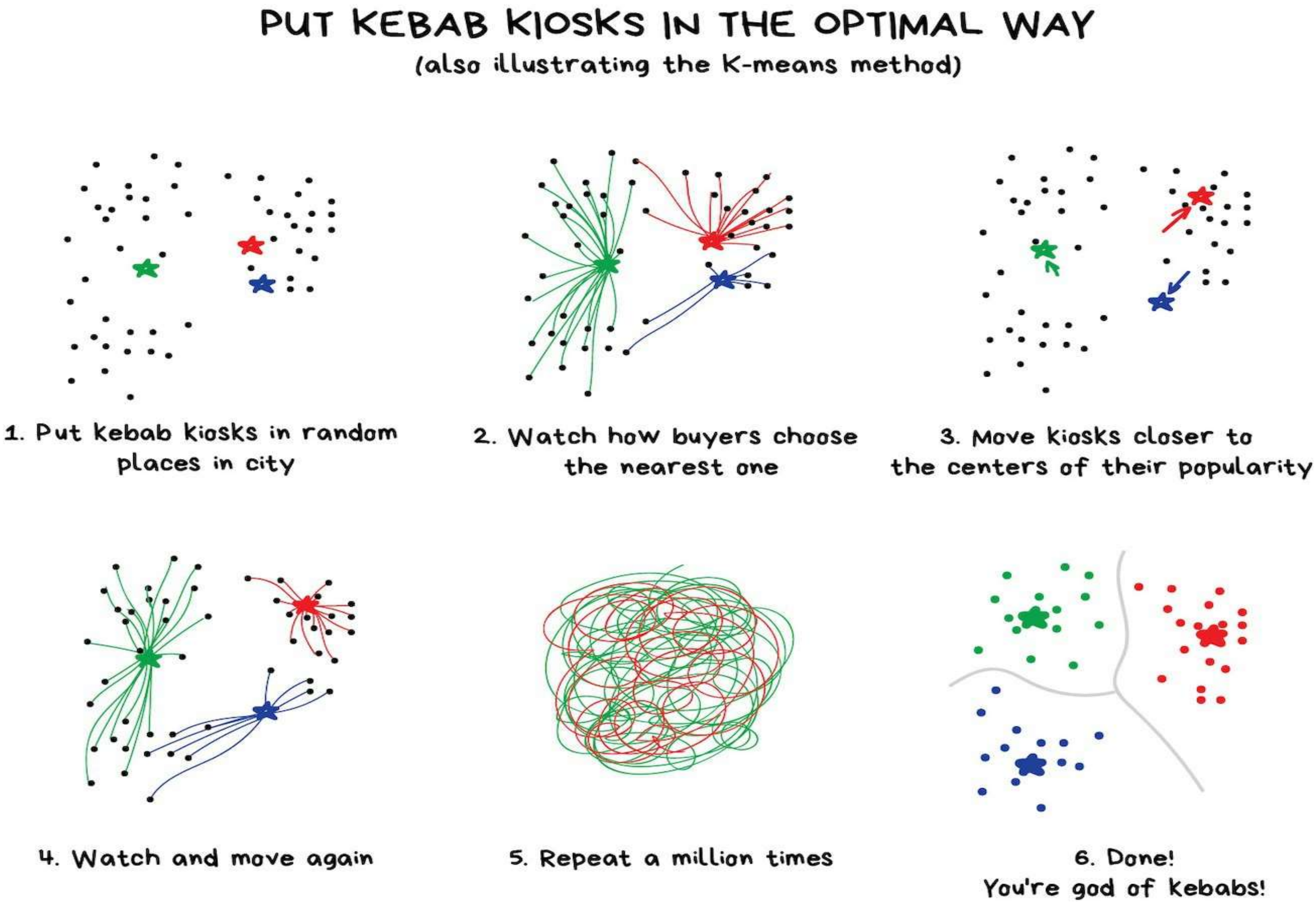
Nowadays used:

- For market segmentation (types of customers, loyalty)
- To merge close points on a map
- For image compression
- To analyze and label new data
- To detect abnormal behavior

Popular algorithms: K-means_clustering, Mean-Shift, DBSCAN.



Unsupervised-Clustering: K-means

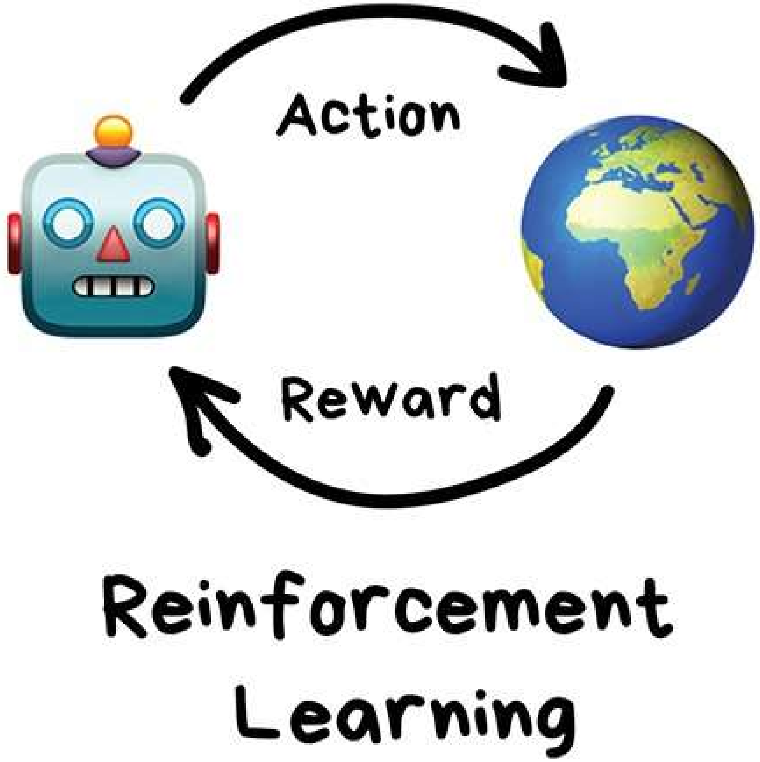


Reinforcement Learning

"Throw a robot into a maze and let it find an exit"

Nowadays used for:
- Self-driving cars
- Robot vacuums
- Games
- Automating trading
- Enterprise resource management

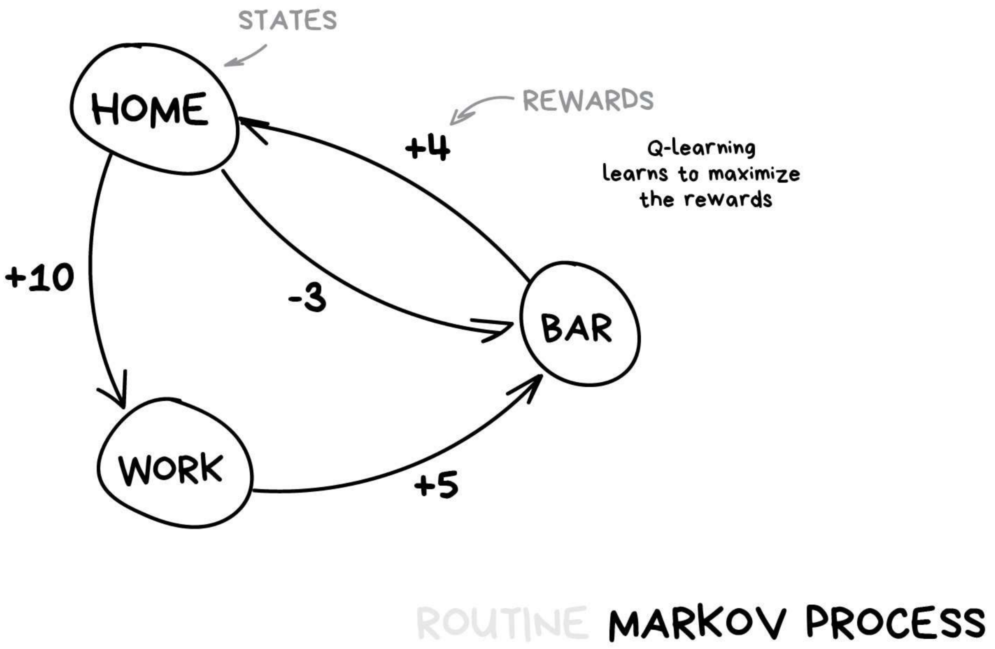
Popular algorithms: Q-Learning, SARSA, DQN, A3C, Genetic algorithm

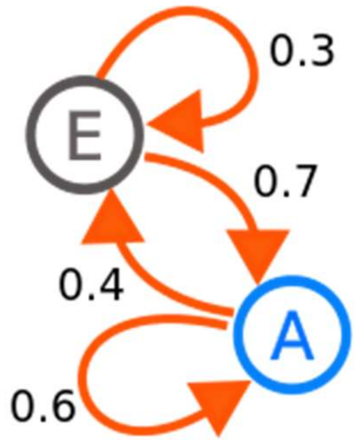


https://youtu.be/qv6UVOQ0F44

Reinforcement Learning: Q-Learning

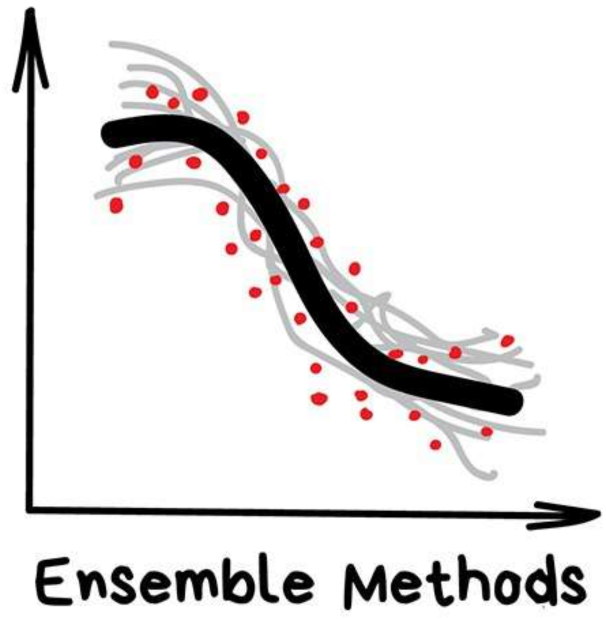
Q-learning and its derivatives (SARSA & DQN). 'Q' in the name stands for "Quality" as a robot learns to perform the most "qualitative" action in each situation and all the situations are memorized as a simple markovian process. A **Markov chain** is a model describing a sequence of possible events in which the probability of each event depends only on the state attained in the previous event.





Ensemble Methods

Ensemble methods use multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone.

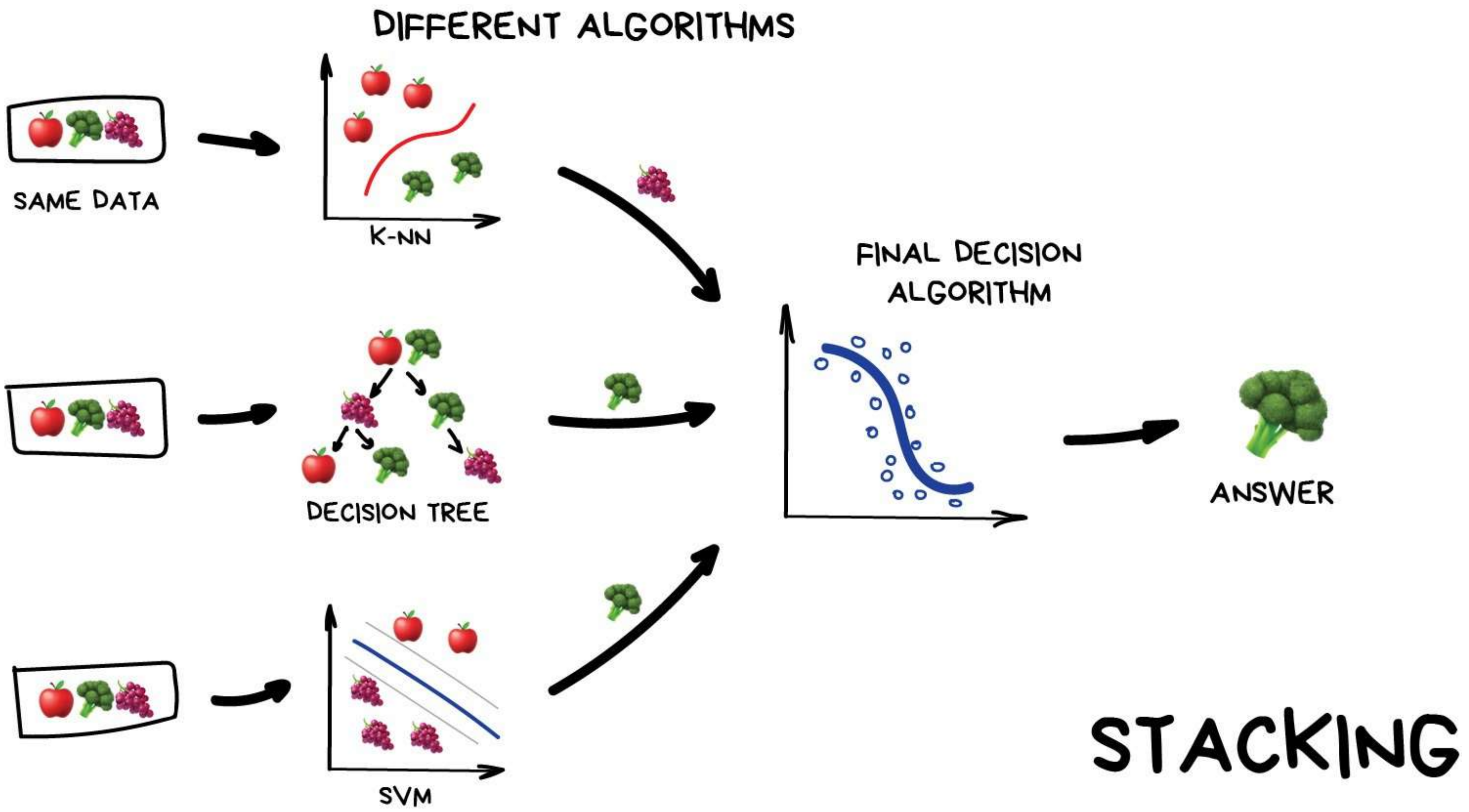


Nowadays is used for:

- Everything that fits classical algorithm approaches (but works better)
- Search systems (★)
- Computer vision
- Object detection

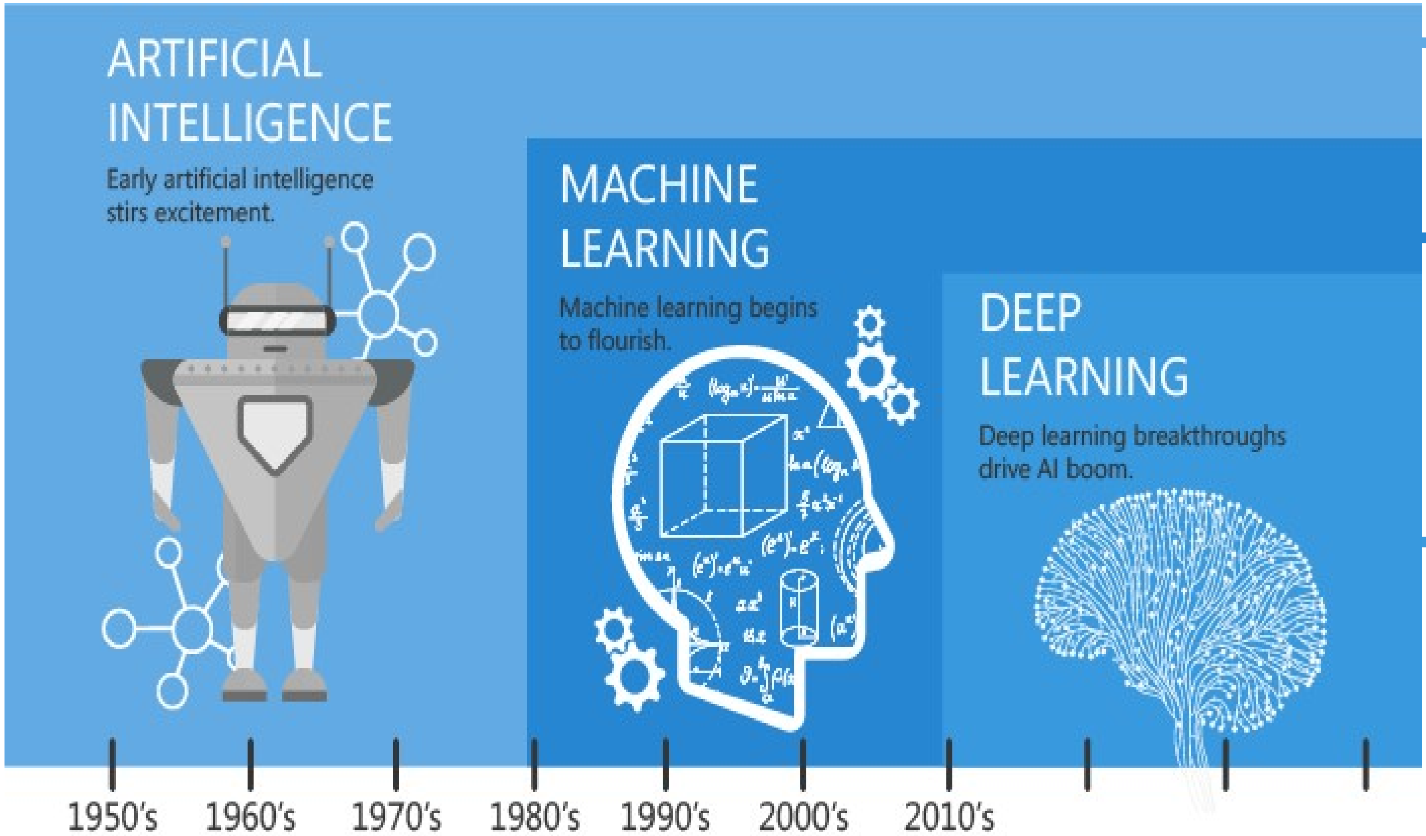
Popular algorithms: Random Forest, Gradient Boosting

Ensemble: Stacking



Like that you ask your friends whether to meet a professor in order to make the final decision of specialization yourself.

Development of Artificial intelligence



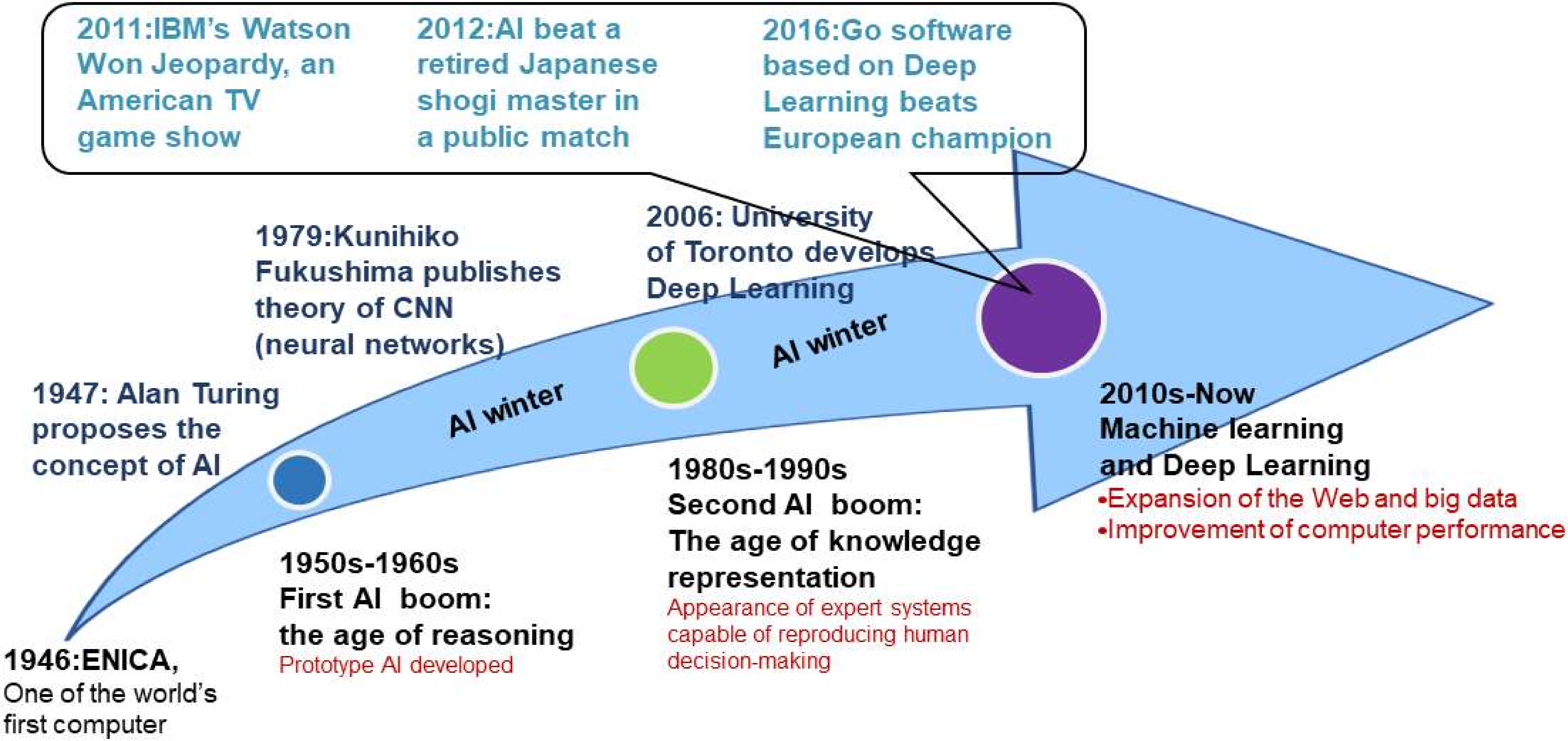
Any technique that enables computers to mimic human intelligence

The subset of AI that includes abstruse statistical techniques that enable machines to improve at tasks with experience.

The subset of machine learning composed of algorithms that permit software to train itself to perform tasks, like speech and image recognition , by exposing multilayered neural networks to vast amounts of data.

Deep Learning

AI Research and Development Timeline



History of Deep Learning

1960s

- **Perceptron (Linear model)**

-Frank Rosenblatt, Cornell University

Single layer ANN with one input layer, one output layer and one hidden layer

1980s

- **Multi-Layer Perceptron**

[Do not have difference with DNN today]

- **BP (Backpropagation)**

[Usually no more than 3 layers is helpful]

-Over Fitting

-Vanishing Gradients

2000s

- **RBM (Restricted Boltzmann Machines) initialization[1]**

-Fix over fiiting

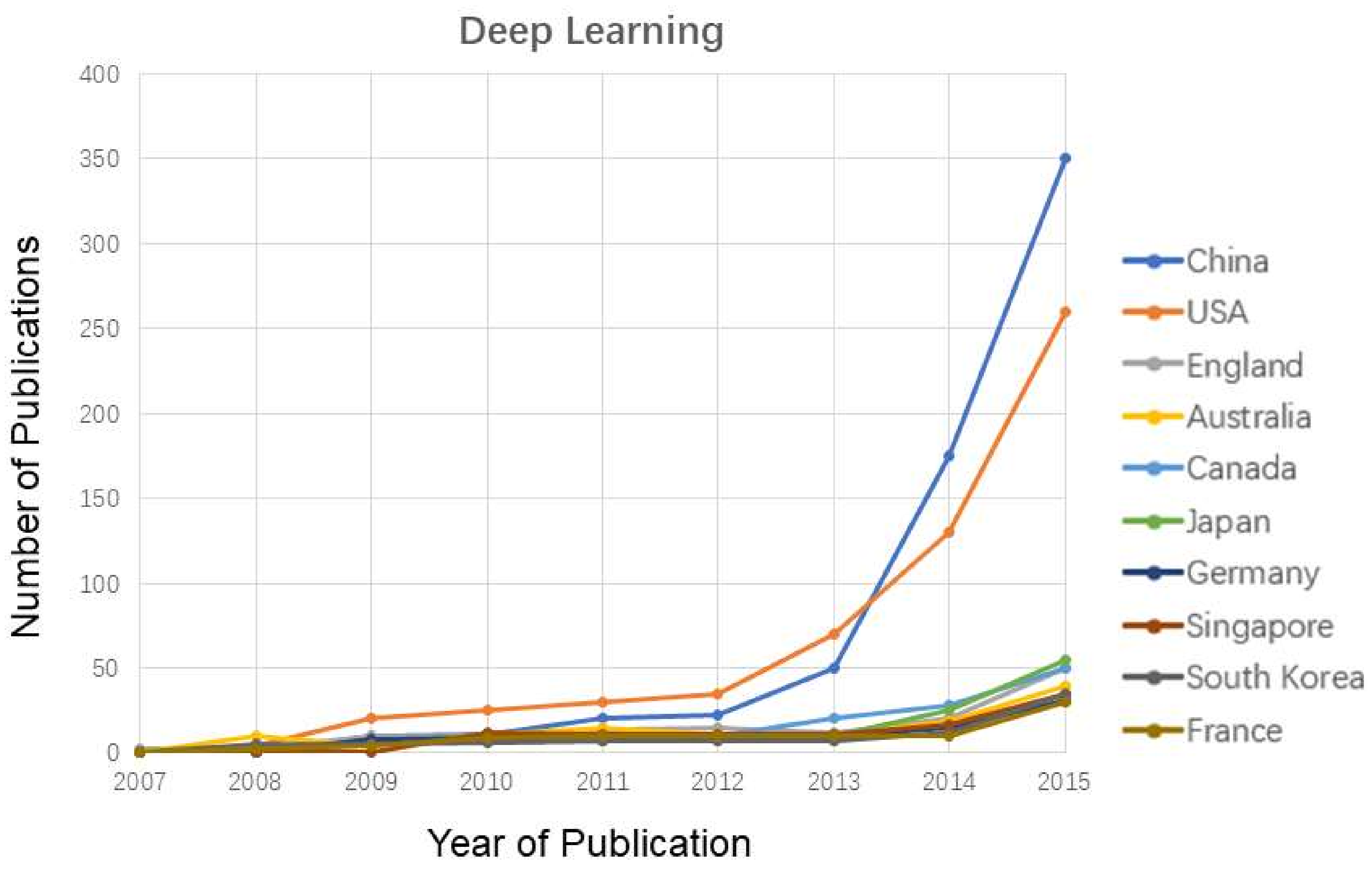
-Hidden layer reaching to 7 layers

- **GPU improvement**

2010s

- **Popular in speech recognition**

[1] Hinton G E, Salakhutdinov R R.Reducing the Dimensionality of Data with Neural Networks[J]. Science, 2006,313(5786):504-507.



Source: China is Now Leading the Race in Artificial Intelligence Development

Machine Learning Tools

	Platform	Cost	Written in language	Algorithms or Features
Scikit Learn	Linux, Mac OS, Windows	Free.	Python, Cython, C, C++	Classification Regression Clustering Preprocessing Model Selection Dimensionality reduction.
PyTorch	Linux, Mac OS, Windows	Free	Python, C++, CUDA	Autograd Module Optim Module nn Module
TensorFlow	Linux, Mac OS, Windows	Free	Python, C++, CUDA	Provides a library for dataflow programming.

Machine Learning Tools

	Platform	Cost	Written in language	Algorithms or Features
Weka	Linux, Mac OS, Windows	Free	Java	Data preparation Classification Regression Clustering Visualization Association rules mining
KNIME	Linux, Mac OS, Windows	Free	Java	Can work with large data volume. Supports text mining & image mining through plugins
Colab	Cloud Service	Free	-	Supports libraries of PyTorch, Keras, TensorFlow, and OpenCV

Machine Learning Tools

	Platform	Cost	Written in language	Algorithms or Features
Apache Mahout	Cross-platform	Free	Java Scala	Preprocessors Regression Clustering Recommenders Distributed Linear Algebra.
Accors.Net	Cross-platform	Free	C#	Classification Regression Distribution Clustering Hypothesis Tests & Kernel Methods Image, Audio & Signal. & Vision

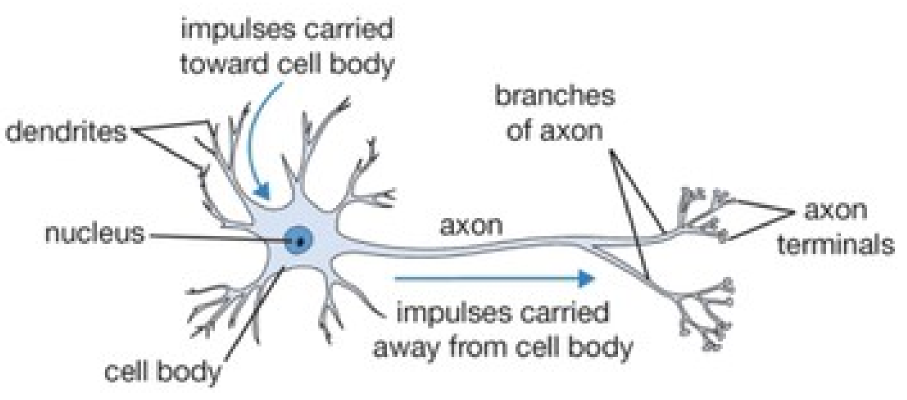
Machine Learning Tools

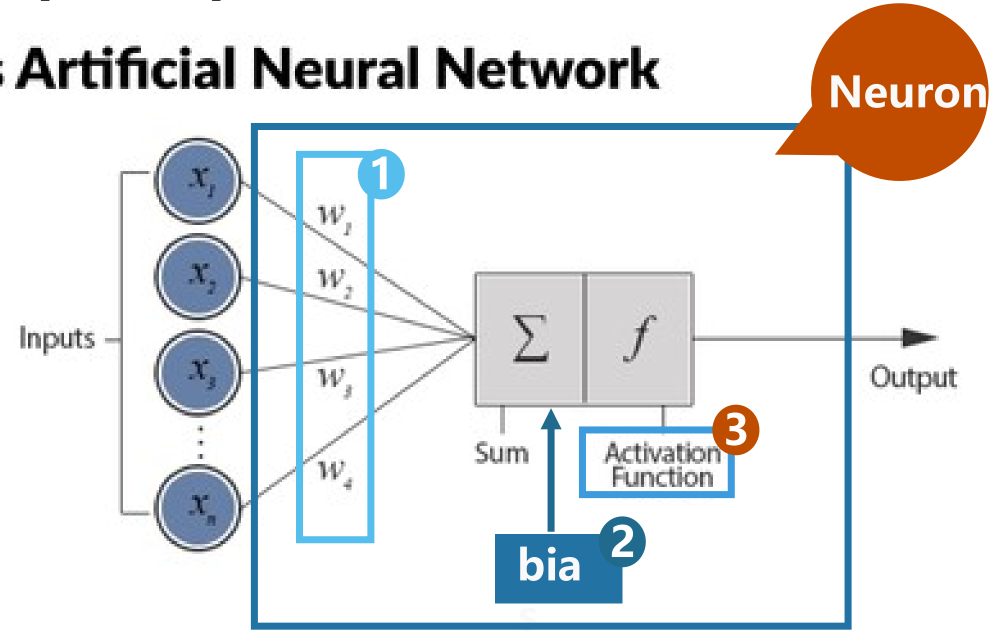
南方科技大学 SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY | 电子与电气工程系 DEPARTMENT OF ELECTRONIC AND ELECTRICAL ENGINEERING

	Platform	Cost	Written in language	Algorithms or Features
Shogun	Windows Linux UNIX Mac OS	Free	C++	Regression Classification Clustering Support vector machines. Dimensionality reduction Online learning etc.
Keras.io	Cross-platform	Free	Python	API for neural networks
Rapid Miner	Cross-platform	Free plan Small: $2500 per year. Medium: $5000 per year. Large: $10000 per year.	Java	Data loading & Transformation Data preprocessing & visualization.

Artificial Neural Network (ANN)

Biological Neuron versus Artificial Neural Network



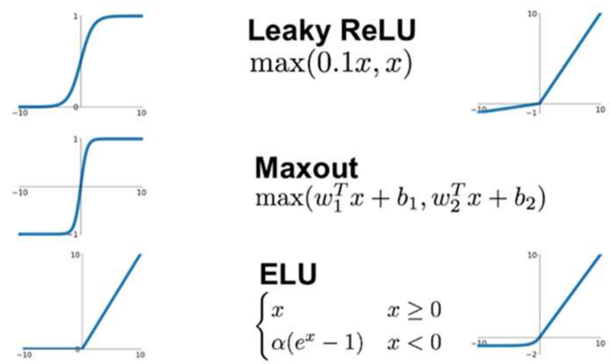


- Neuron = Logical function
- Network Parameter : all the weights[1] and biases[2] in the "neurons"

1. Weight: each connection between two neurons has a weight w.
2. Bias: view as threshold, is an "extra" neuron added to each pre-output layer

* Weight and bias depends on the training data

3. Activation Functions



Sigmoid: $\sigma(x) = \frac{1}{1+e^{-x}}$

tanh: $\tanh(x)$

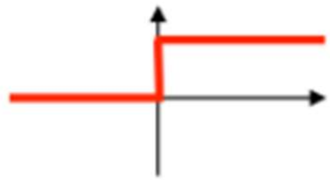
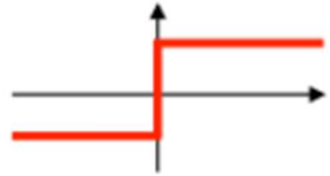
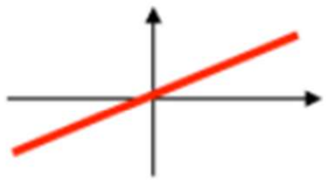
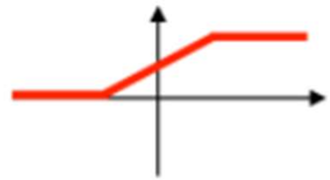
ReLU: $\max(0, x)$

Leaky ReLU: $\max(0.1x, x)$

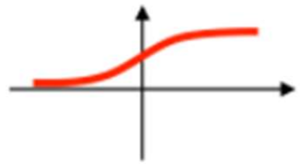
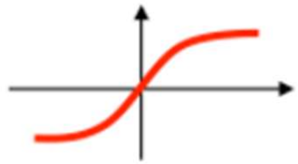
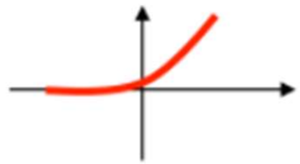
Maxout: $\max(w_1^T x + b_1, w_2^T x + b_2)$

ELU: $\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$

Activation Functions for ANN

Activation function	Equation	Example	1D Graph
Unit step (Heaviside)	$\phi(z) = \begin{cases} 0, & z < 0, \\ 0.5, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Sign (Signum)	$\phi(z) = \begin{cases} -1, & z < 0, \\ 0, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Linear	$\phi(z) = z$	Adaline, linear regression	
Piece-wise linear	$\phi(z) = \begin{cases} 1, & z \geq \frac{1}{2}, \\ z + \frac{1}{2}, & -\frac{1}{2} < z < \frac{1}{2}, \\ 0, & z \leq -\frac{1}{2}, \end{cases}$	Support vector machine	

Activation Functions for ANN

Activation function	Equation	Example	1D Graph
Logistic (sigmoid)	$\phi(z) = \frac{1}{1+e^{-z}}$	Logistic regression, Multi-layer NN	
Hyperbolic tangent	$\phi(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	Multi-layer Neural Networks	
Rectifier, ReLU (Rectified Linear Unit)	$\phi(z) = max(0, z)$	Multi-layer Neural Networks	
Rectifier, softplus (http://sebastianraschka.com)	$\phi(z) = \ln(1 + e^z)$	Multi-layer Neural Networks	

Three steps of Deep Learning
(Deep Neural Network)

Step 1: Network Structure ⇨ **Step 2: Learning Target** ⇨ **Step 3: Learn!**

A set of function

Define the goodness of a function

Pick the best Function f*

Different structures of ANN

- **Neural Network :** Different connections leads to different network structures

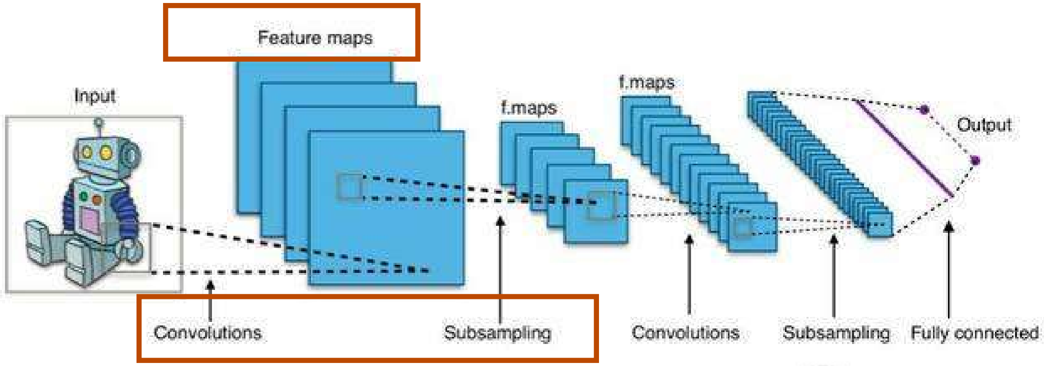
CNN

- **Convolution Neural Network**

Image Processing

Feature engineering/extraction

Feed-forward Neural Networks



RNN

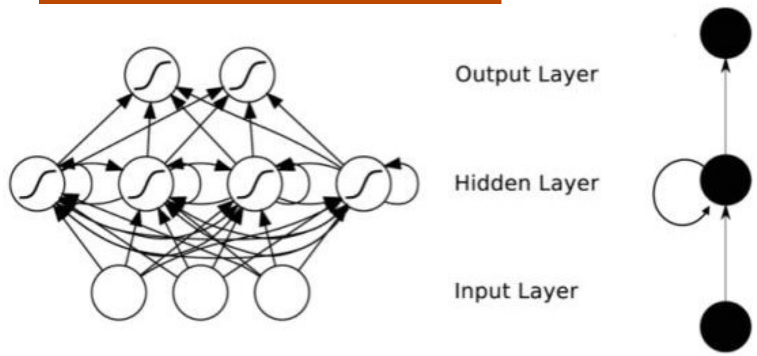
- **Recurrent Neural Network**

Natural Language Processing

Adv./Dis. Temporal dynamic

Updated vision: LSTM (Long short-term memory)

vs

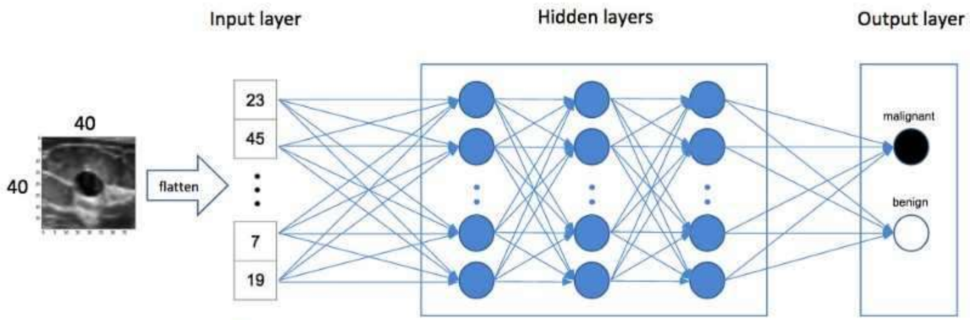


DNN

- **Deep Neural Network (Deep Learning)**

Adv. Without pre-feature extracting

Dis. Computing power demanding



GAN (Generative Adversative Nets)

- A discriminative model learns a function that maps the input data (x) to some desired output class label (y). In probabilistic terms, they directly learn the conditional distribution P(y|x).

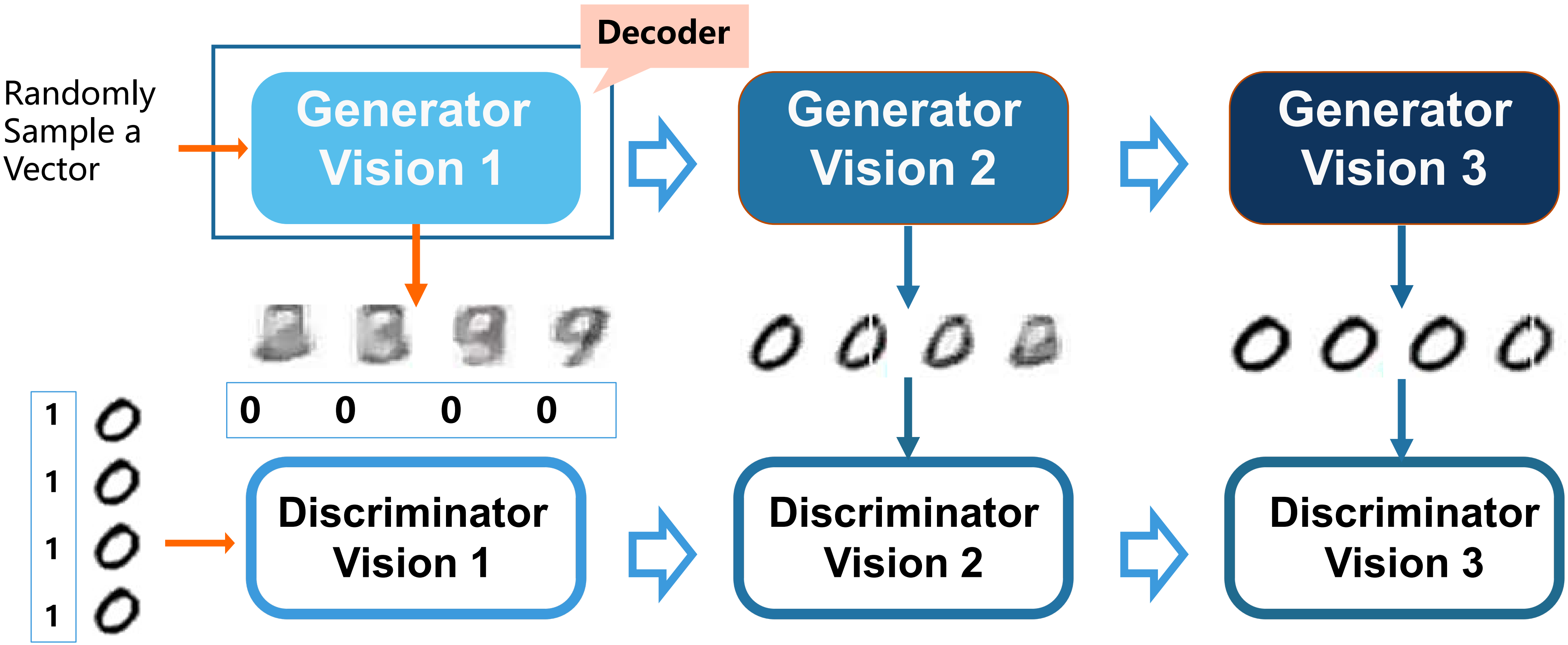
- A generative model tries to learn the joint probability of the input data and labels simultaneously, i.e. P(x,y). This can be converted to P(y|x) for classification via Bayes rule, but the generative ability could be used for something else as well, such as creating likely new (x, y) samples.

GAN (Generative Adversative Nets)

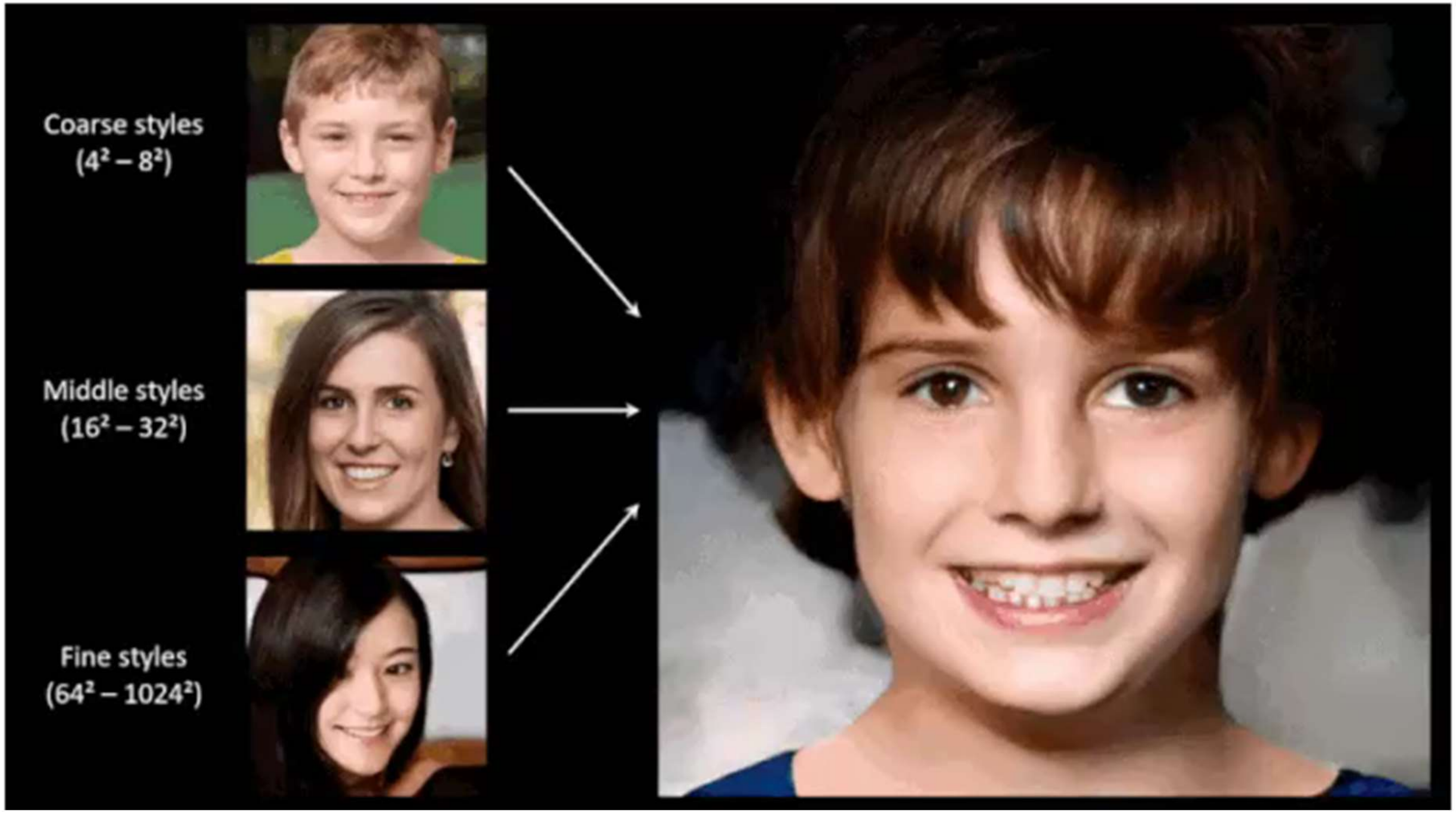
GAN was first introduced in 2014 by a group of researchers at the University of Montreal lead by Ian Goodfellow.

The main idea behind a GAN is to have two competing neural network models. One takes noise as input and generates samples (and so is called the generator). The other model (called the discriminator) receives samples from both the generator and the training data, and has to be able to distinguish between the two sources. These two networks play a continuous game, where the generator is learning to produce more and more realistic samples, and the discriminator is learning to get better and better at distinguishing generated data from real data. These two networks are trained simultaneously, and the hope is that the competition will drive the generated samples to be indistinguishable from real data.

Evolution of generation



NVIDIA--StyleGAN



https://www.youtube.com/watch?v=kSLJriaOumA&feature=youtu.be

Tero Karras, Samuli Laine, Timo Aila, A Style-Based Generator Architecture for Generative Adversarial Networks[J], 6 Feb 2019

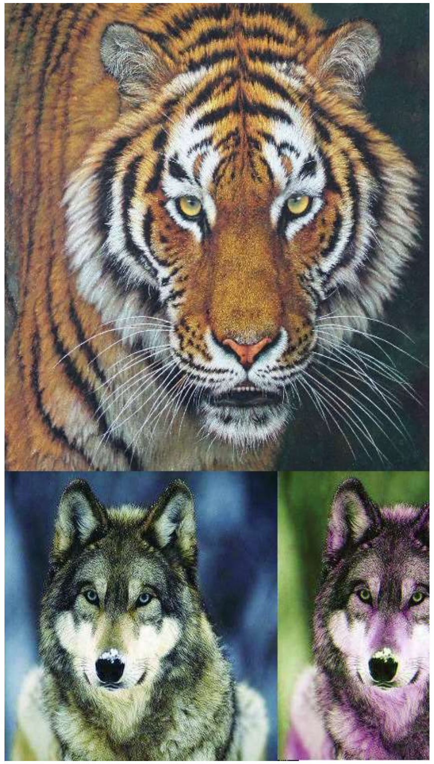
Transfer Learning

Transfer Learning

Task (Target data)
Cat and Dog Classifier

Source Data

Same Input distribution
Different task

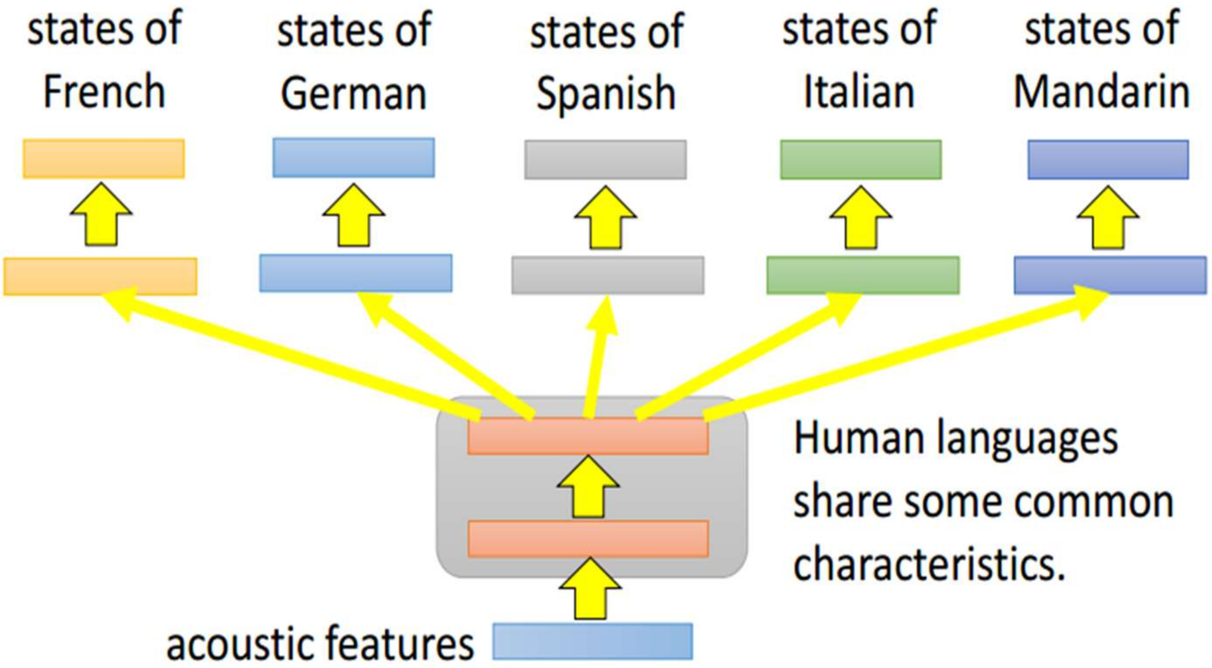


Same Task
Different Input distribution

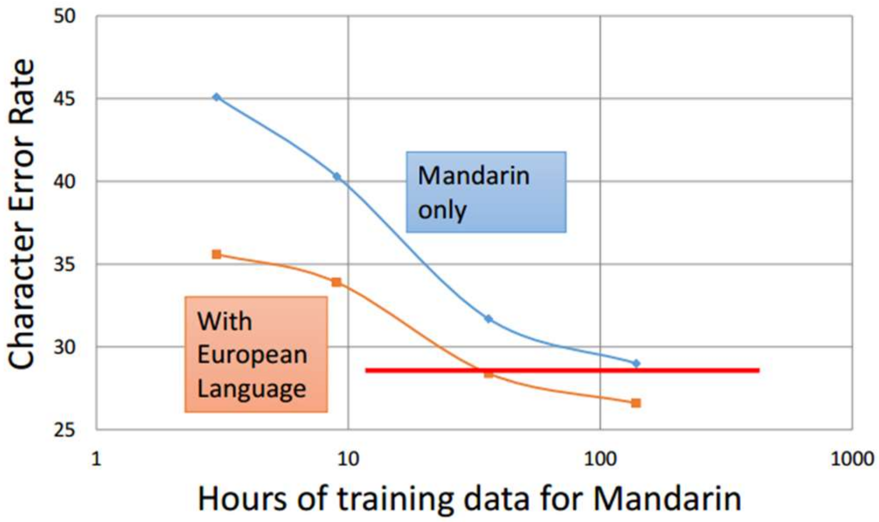
Multitask Learning

- **The multi-layer structure makes NN suitable for multitask learning**

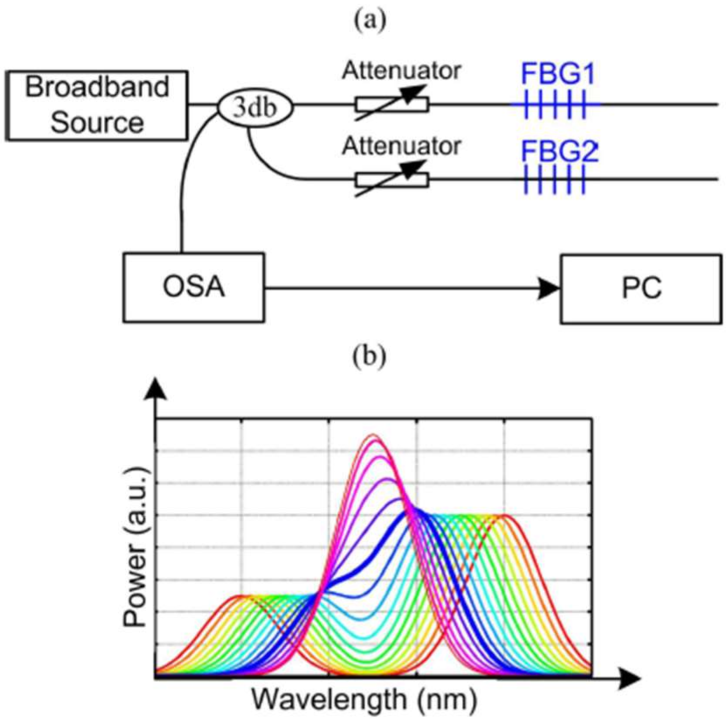
Example: Multilingual Speech Recognition

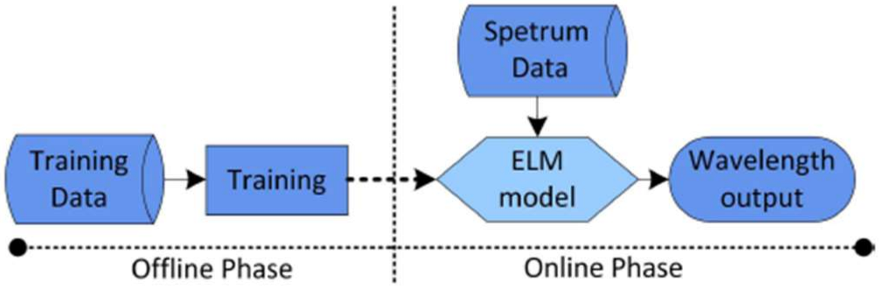


Human languages share some common characteristics.



Wavelength Detection in Spectrally Overlapped FBG Sensor Network Using ELM





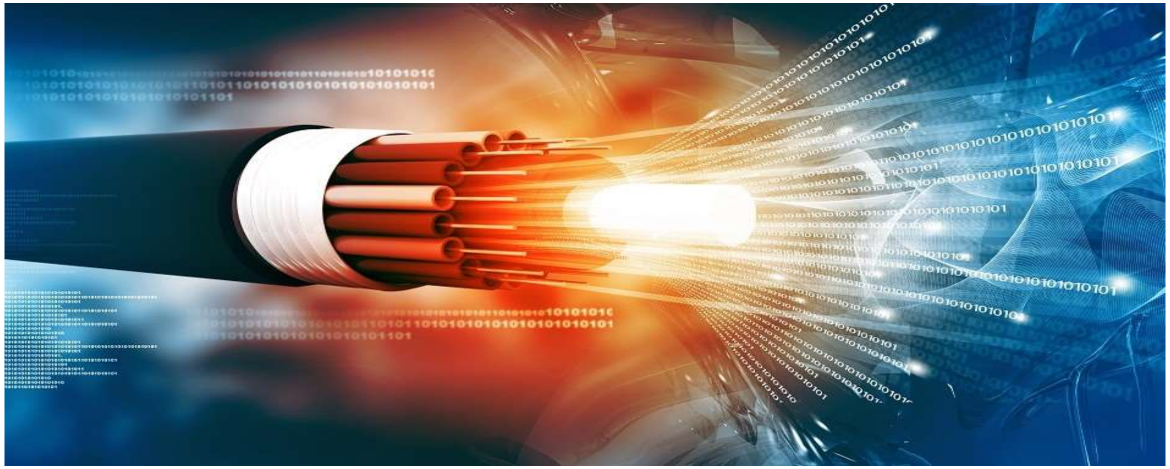
Methods	RMS (pm)	Training Time (s)	Testing Time (s)
ELM	0.918	308	0.215
LS-SVR	1.206	17290	0.578
DE	1.912	-	82.724
PSO	2.403	-	168.92

Note: ”-” means there is no training time.

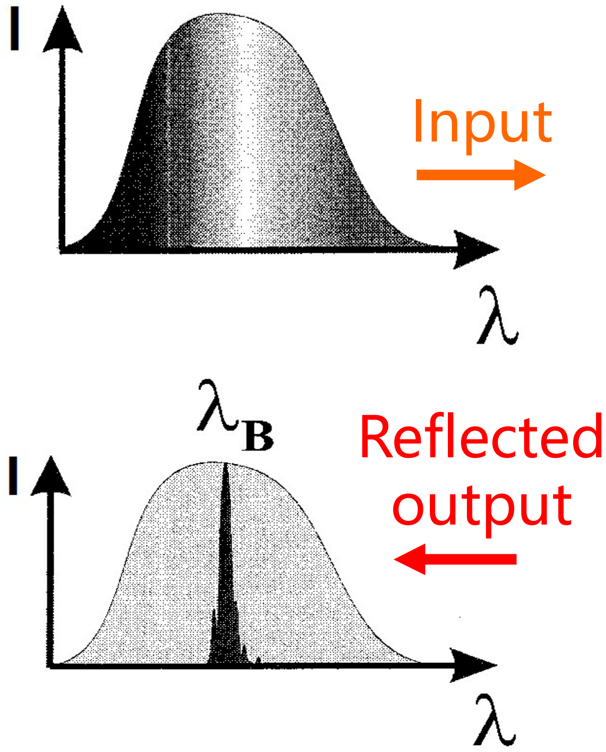
LS-SVR: Least Squares Support Vector Regression
DE: Differential Evolution
PSO: Particle Swarm Optimizer

Jiang, H., Chen, J., & Liu, T. (2014). Wavelength detection in spectrally overlapped FBG sensor network using extreme learning machine. *IEEE Photonics Technology Letters*, *26*(20), 2031-2034.

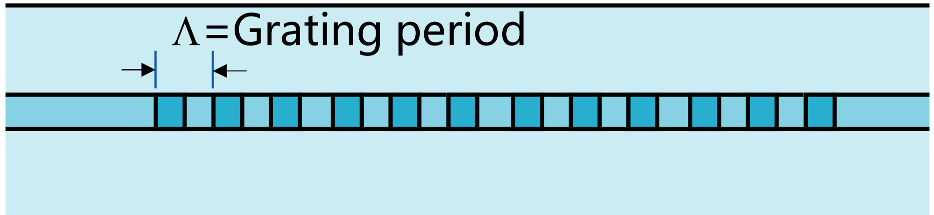
AI Photonics

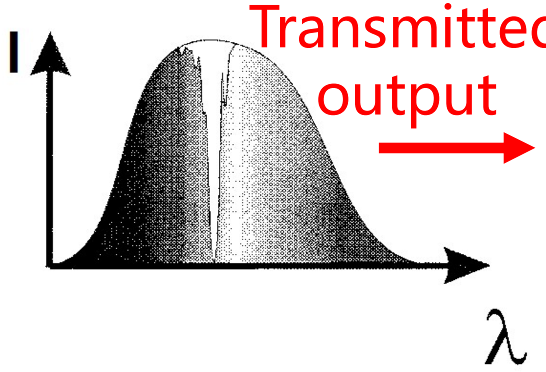


Introduction: Fiber Bragg Grating (FBG)



All-silica, glass fiber with grating written in core





Bragg Condition: $\lambda_B = 2n_{eff}\Lambda$

- ✓ Typical strain response ~1 pm /με
- ✓ Typical temperature response ~10 pm/℃

Introduction: Fiber Bragg Grating (FBG)

Multi-parameters monitoring (Temperature and Strain)

$$\Delta\lambda_B = 2n\Lambda\left(\left\{1-\underbrace{\boxed{\left(\frac{n^2}{2}\right)[P_{12}-v(P_{11}+P_{12})]}}_{\sim 0.22}\right\}\varepsilon + \left[\underbrace{\boxed{\alpha+\frac{\left(\frac{\mathrm{d}n}{\mathrm{d}T}\right)}{n}}}_{\sim 6.67\times 10^{-6}}\right]\Delta T\right)$$

Under steady Temperature

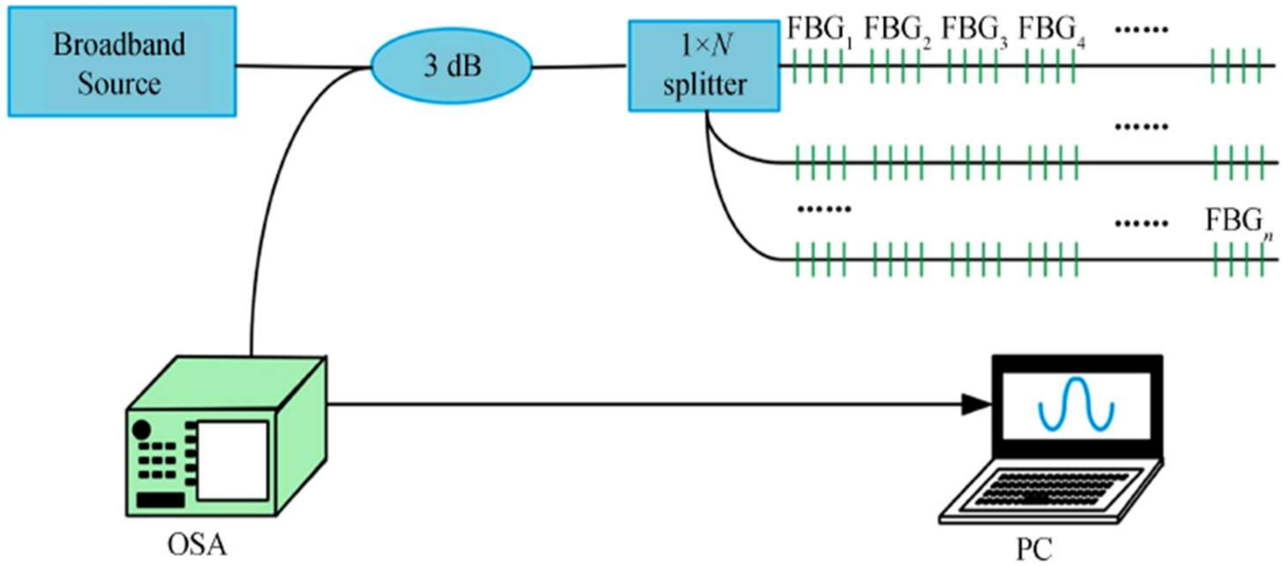
$$\frac{1}{\lambda_B}\frac{\delta\lambda_B}{\delta\varepsilon} = 0.78\times 10^{-6}\mu\varepsilon^{-1}$$

Under steady Strain

$$\frac{1}{\lambda_B}\frac{\delta\lambda_B}{\delta T} = 6.67\times 10^{-6}{}^{\circ}\mathrm{C}^{-1}$$

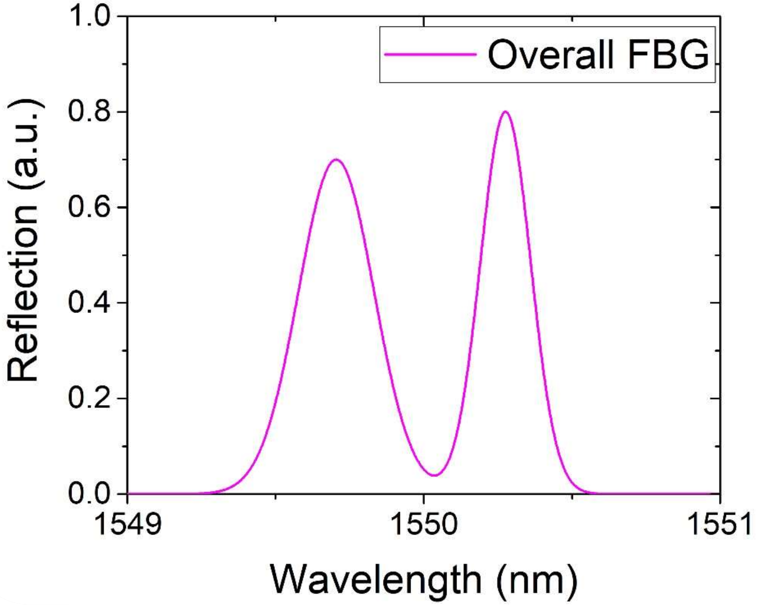
This implies a wavelength resolution of ~1 pm is required for signal demodulation for 1 $\mu\varepsilon$ and ~0.1℃

Introduction: Quasi-distributed FBG sensor network

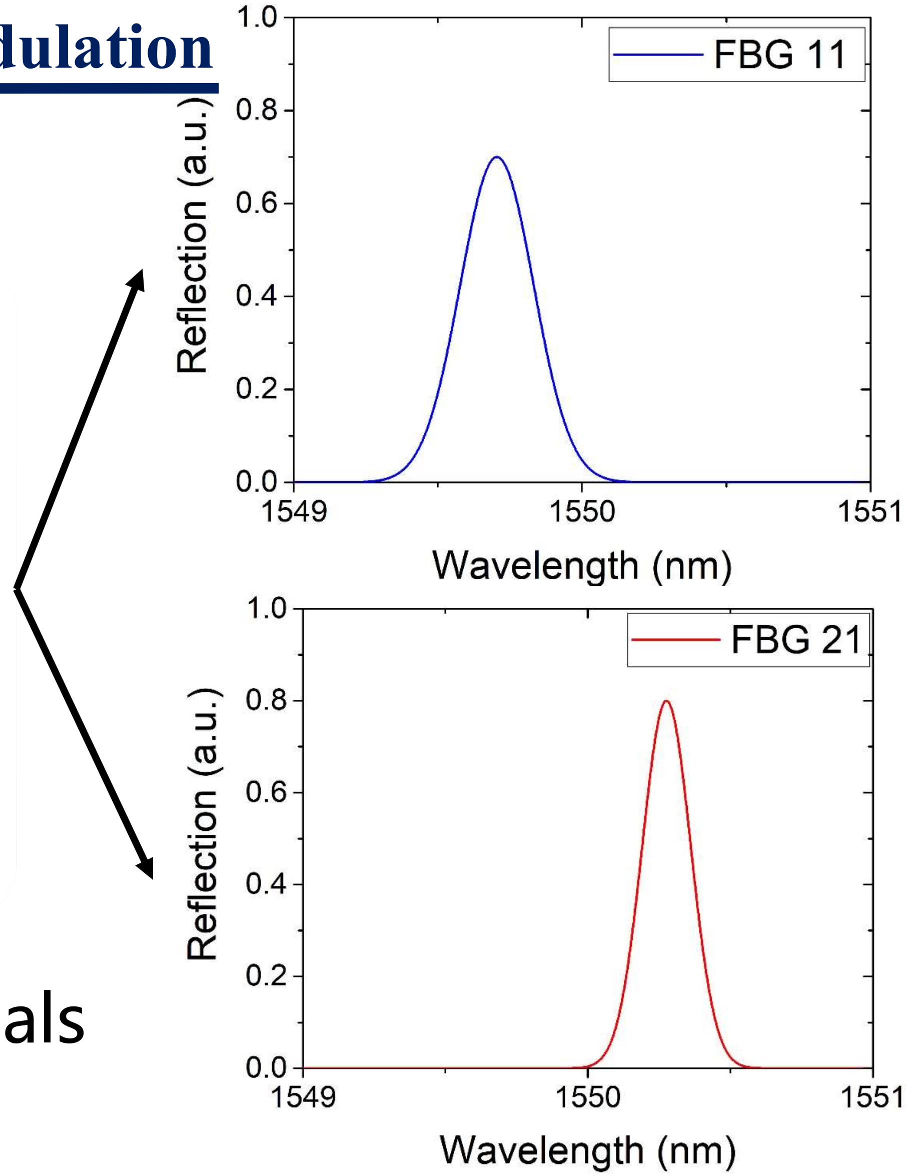


Introduction: Signal demodulation

Input



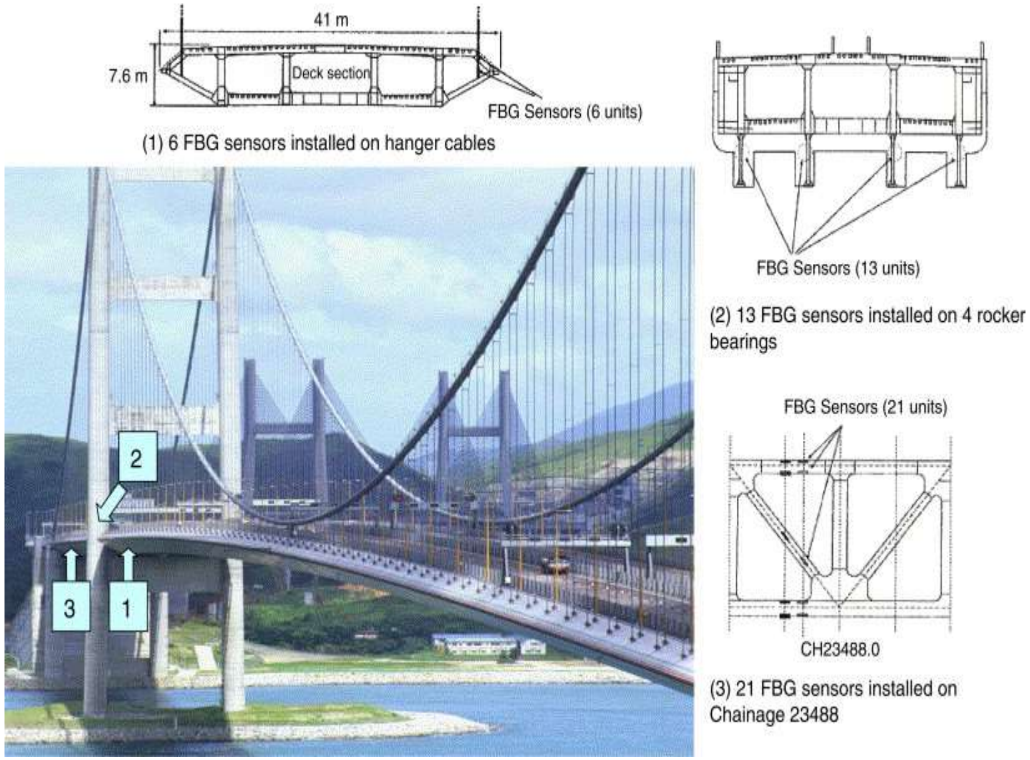
Outputs



- Output the individual signals

- Input a sum of many signals (peaks)

Introduction: Important applications



- FBG under MRT for multi-point leakage monitoring

- In-fence perimeter sensor by embedding FBG in wire fencing

Fiber Bragg grating sensors for structural health monitoring of Tsing Ma bridge: Background and experimental observation, Engineering Structures, Volume 28, Issue 5,2006,Pages 648-659,ISSN 0141-0296

Existing demodulation methods

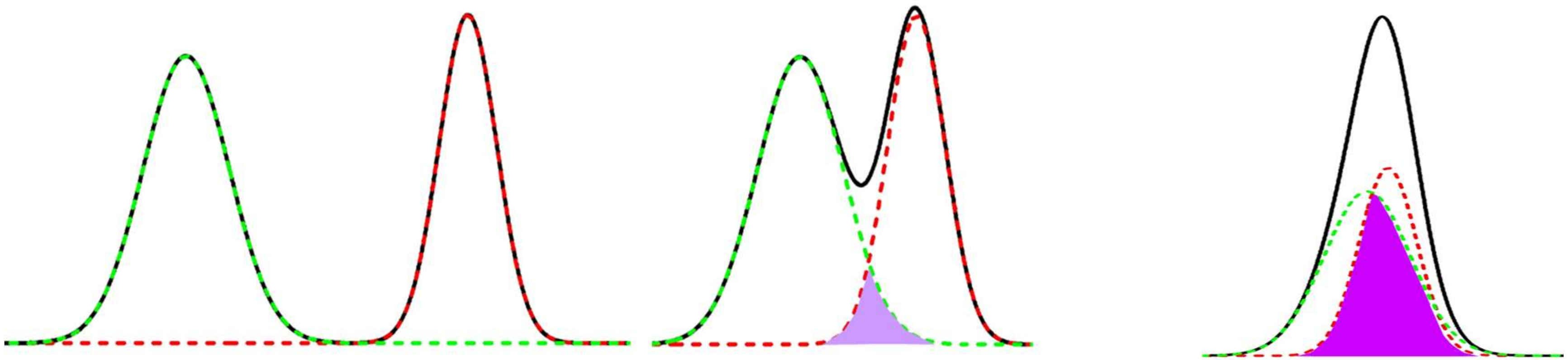
Method	RMS error (pm)	Testing Time (ms)
Particle Swarm Optimizer	2.403	168,920
Differential Evolution	1.912	82,724
Least Square Support Vector Regression	1.206	578
Extremely Learning Machine	0.918	215
Long short-term memory	0.674 (>10 for highly overlapped signal)	1.625

Machine
-learning

Jiang, H., Chen, J., & Liu, T. (2014). Wavelength detection in spectrally overlapped FBG sensor network using extreme learning machine. *IEEE Photonics Technology Letters*, *26*(20), 2031-2034.

Existing demodulation methods

Minimal to moderate overlap: Easy to determine peak frequency



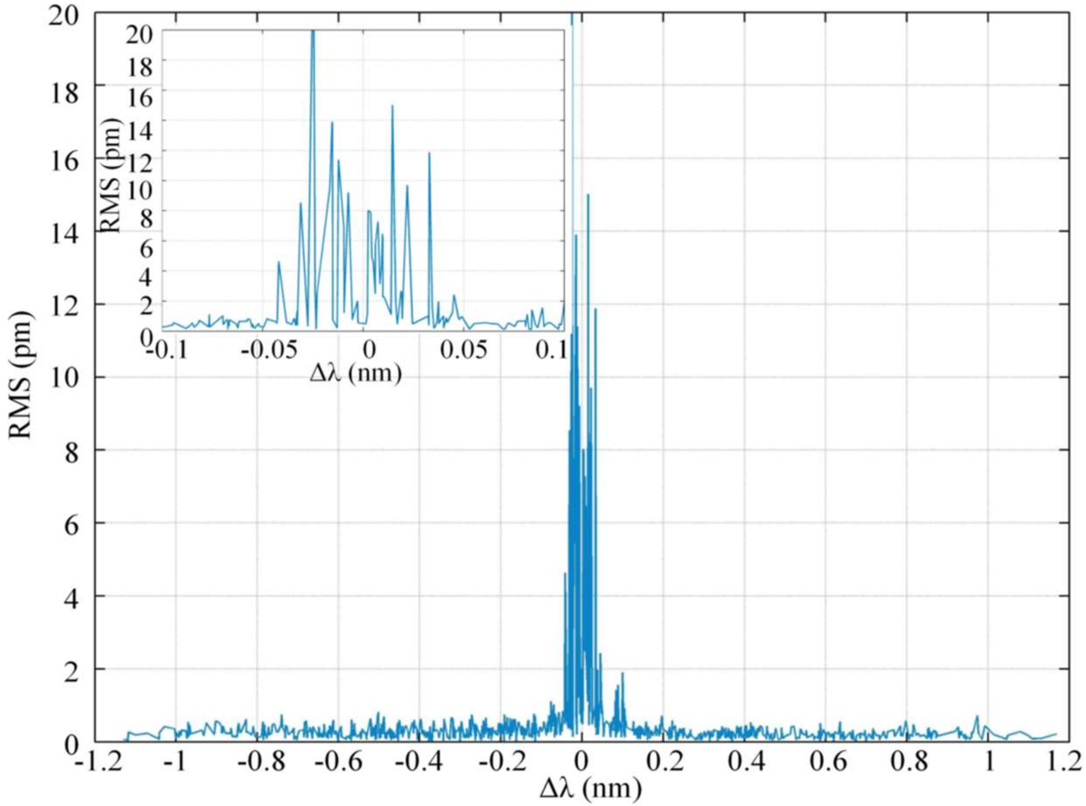
Minimal overlap   Moderate overlap   High overlap

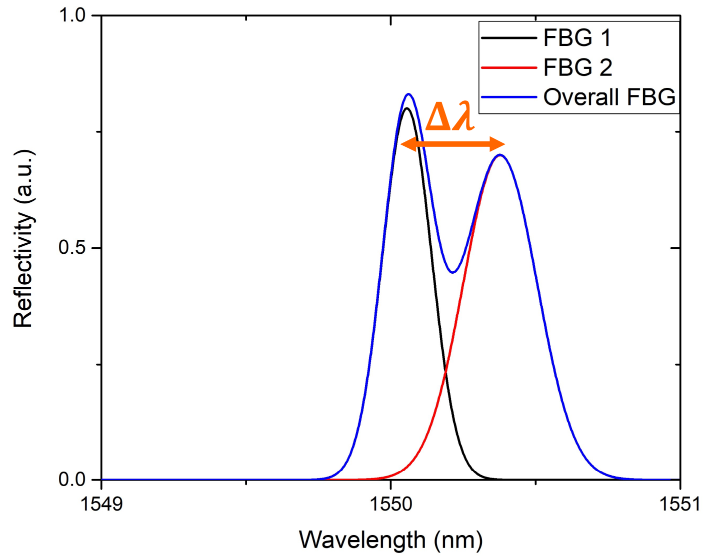
High overlap ($\Delta\lambda < \mathbf{0.05}\boldsymbol{nm}$): Errors spike due to distortion from close peaks

Existing demodulation methods

Long short-term memory (Recurrent Neural Network)

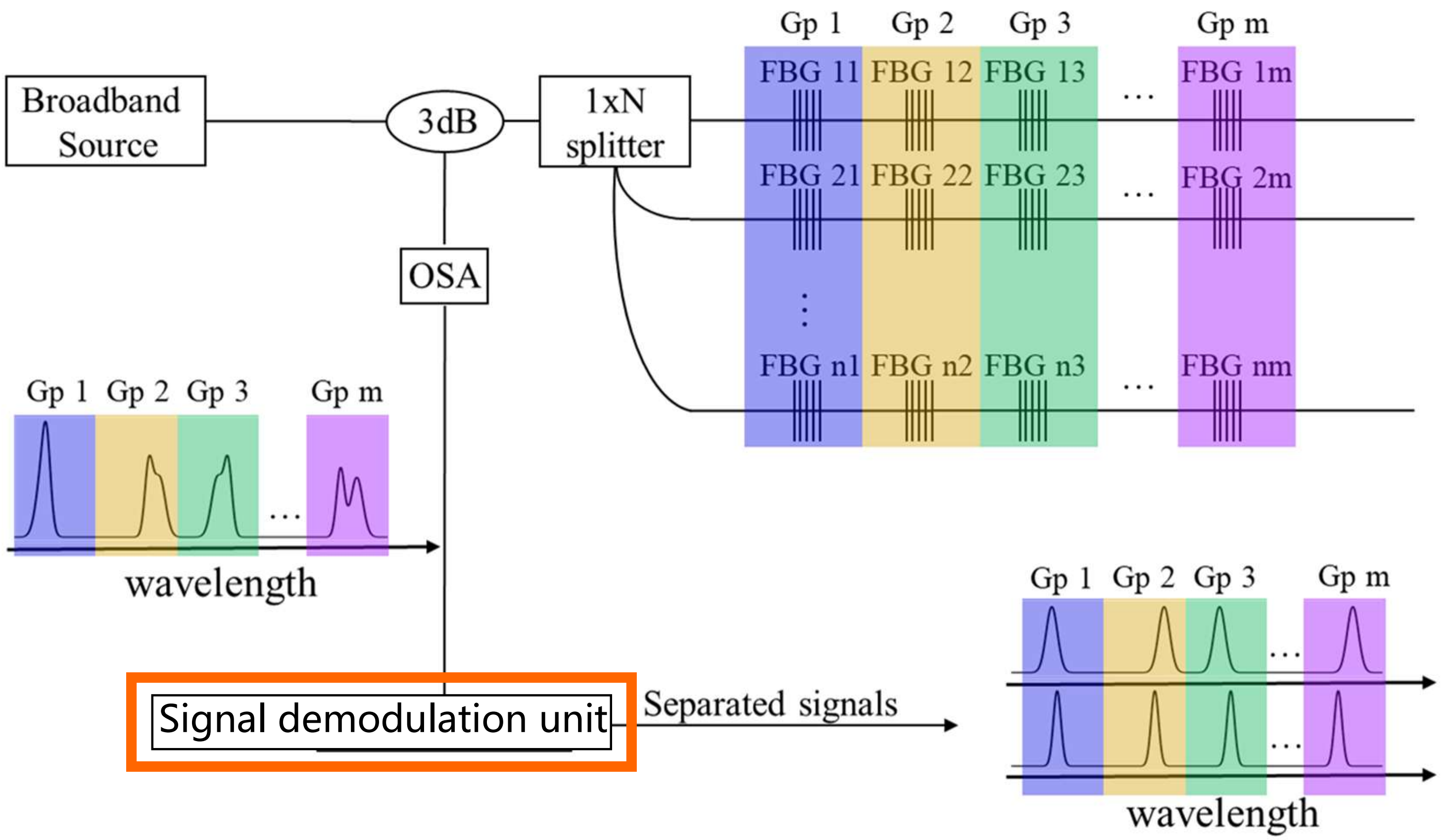
- Error spikes up when the Δλ is small



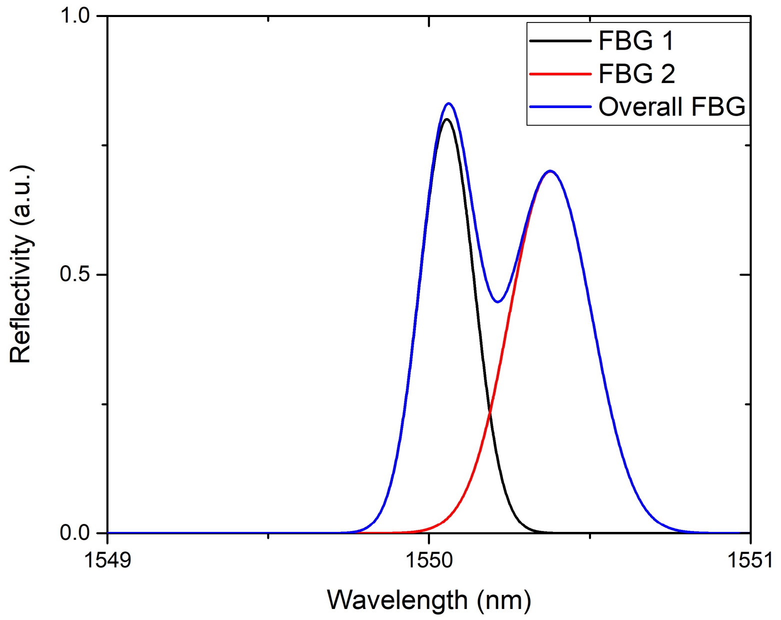


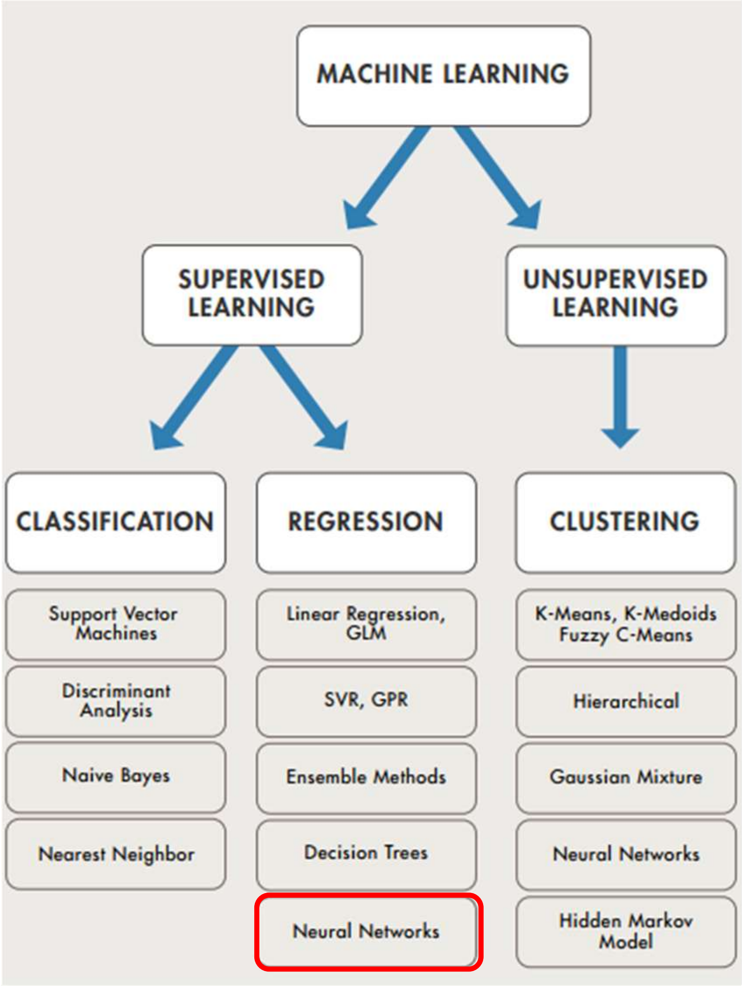
H. Jiang *et al.*, "Wavelength detection of model-sharing fiber Bragg grating sensor networks using long short-term memory neural network," *Optics express,* vol. 27, no. 15, pp. 20583-20596, 2019.

Schematic diagram

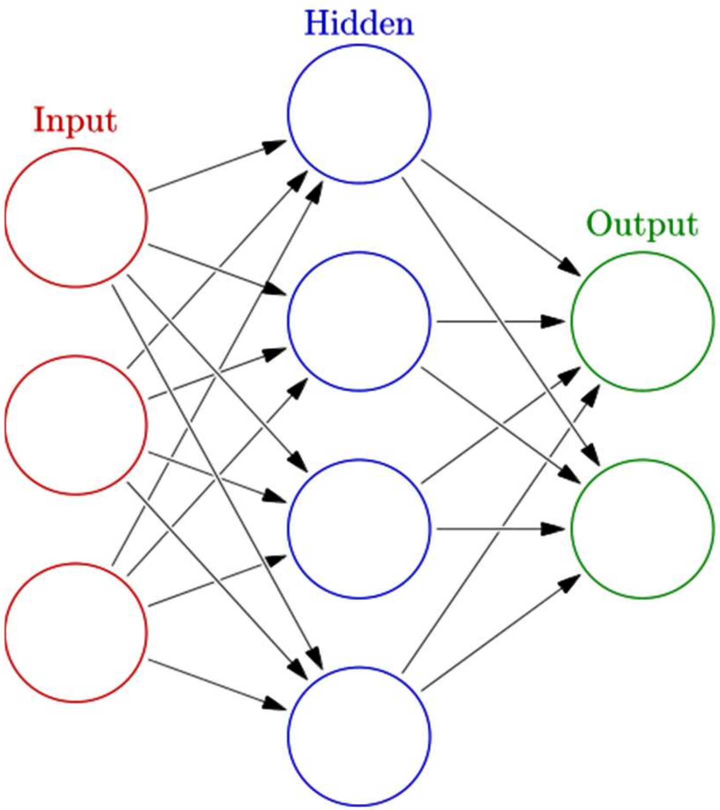


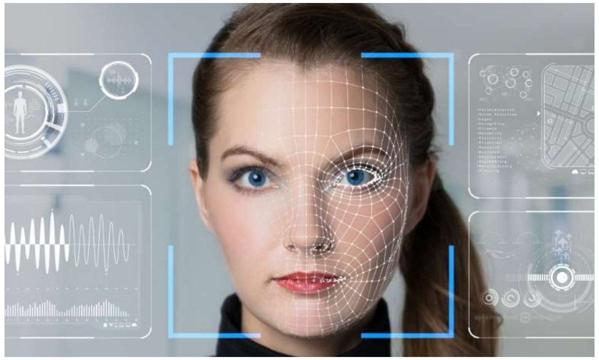
Problem definition





Convolutional Neural Network

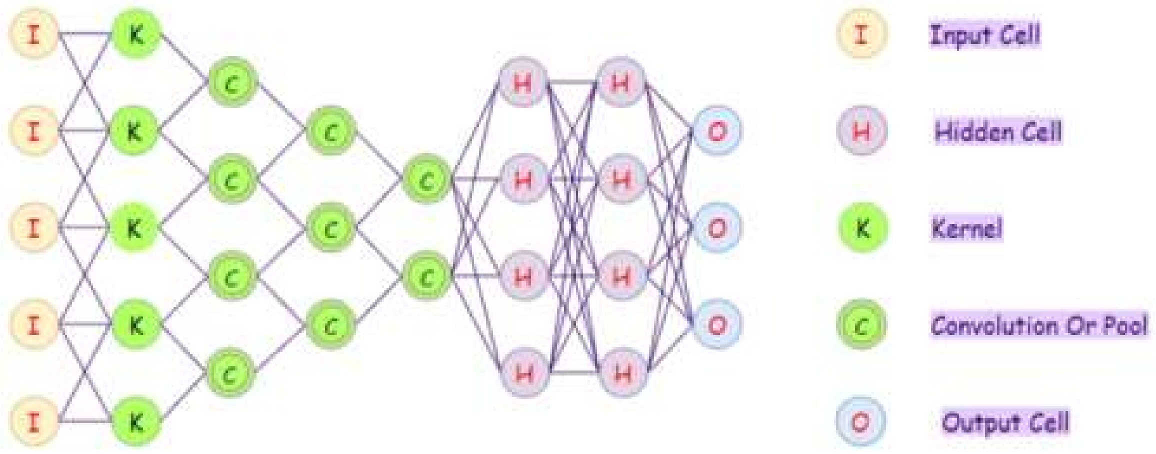


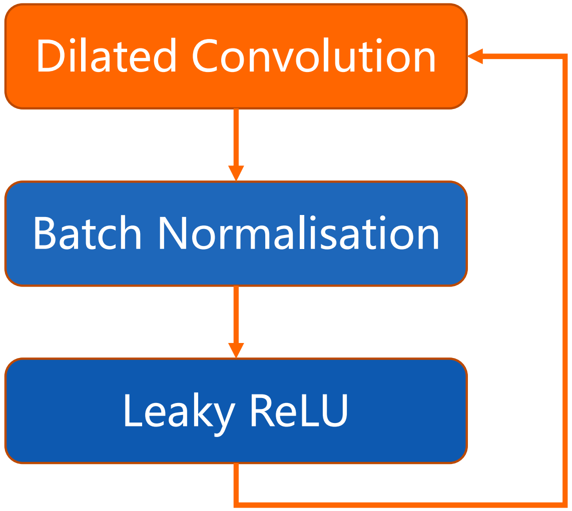


- Normally used for image recognition, anomaly detection etc.
- Only starting to be applied to fiber sensors
- Two common classes:
  - Recurrent Neural Network (RNN)
  - Convolutional Neural Network (CNN)

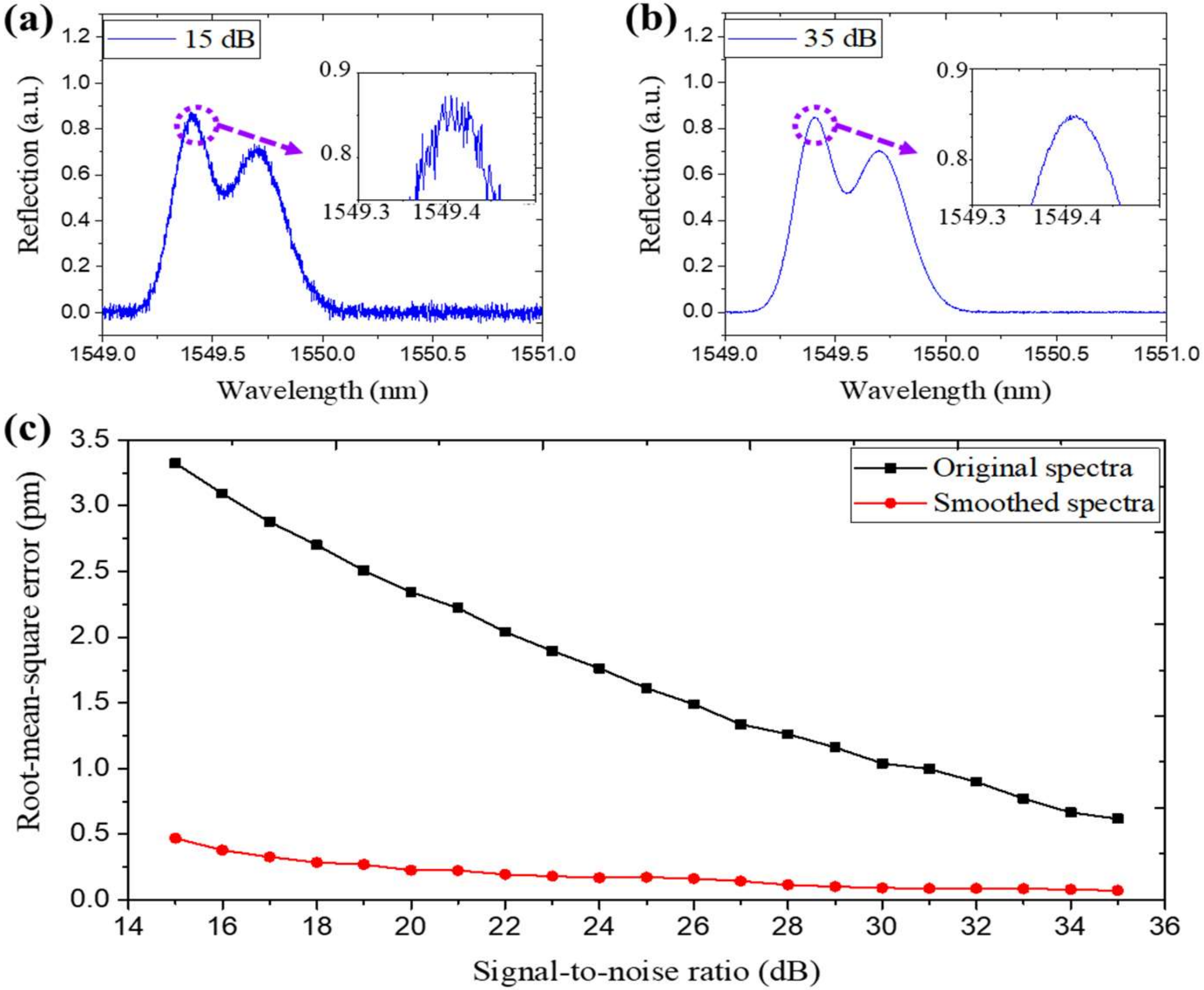
Present work: Convolutional Neural Network

Convolutional neural network





Results (noisy signal)



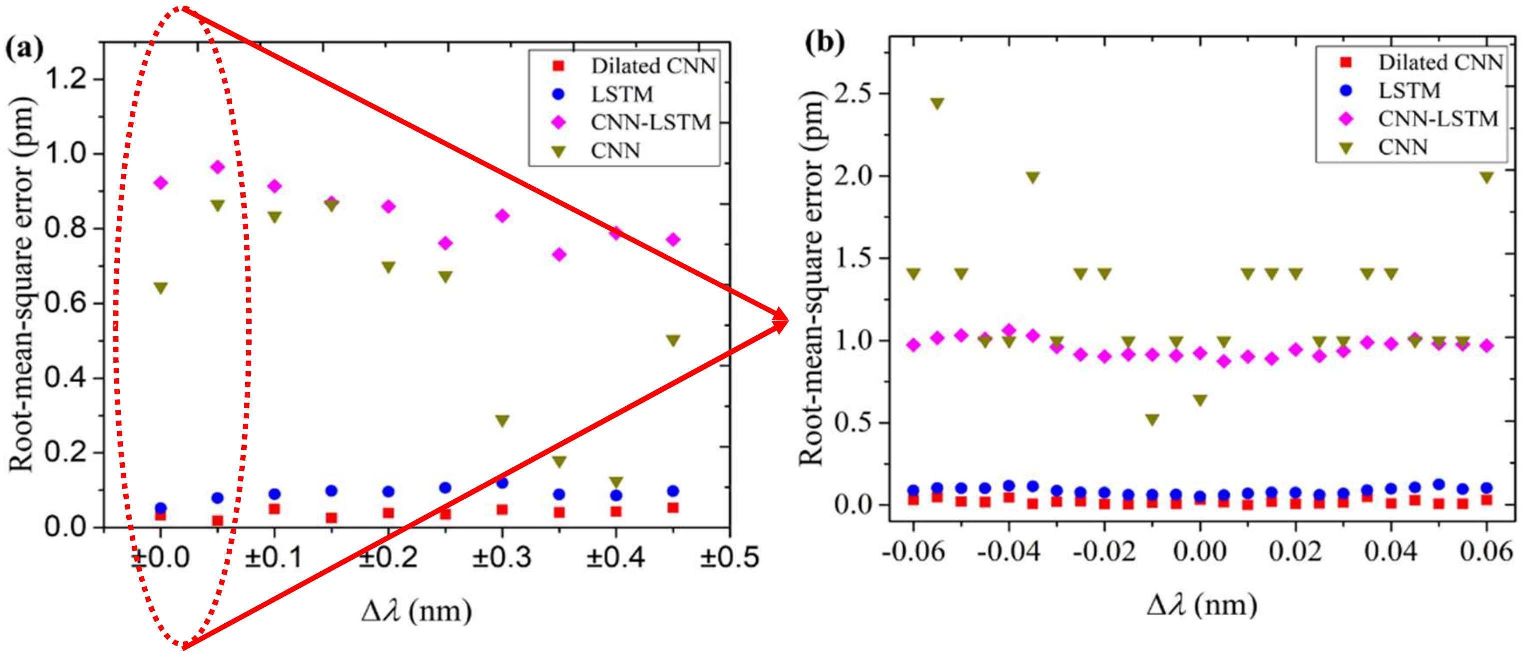
- Gaussian-smoothing filter
- RMS < 0.47 pm when SNR is 15dB
- Robust model that could deal with signal with different SNR

Present work: Comparison table (Published works)

Method	RMS error (pm)	Testing Time (ms)
Particle Swarm Optimizer	2.403	168,920
Differential Evolution	1.912	82,724
Least Square Support Vector Regression	1.206	578
Extremely Learning Machine	0.918	215
Long short-term memory	~0.674 (>10 for high overlap)	1.625
Dilated convolutional neural network	**0.0201**	**15.17**

Present work: Comparison between our models

High overlap region



Present work: Comparison table (models built by us)

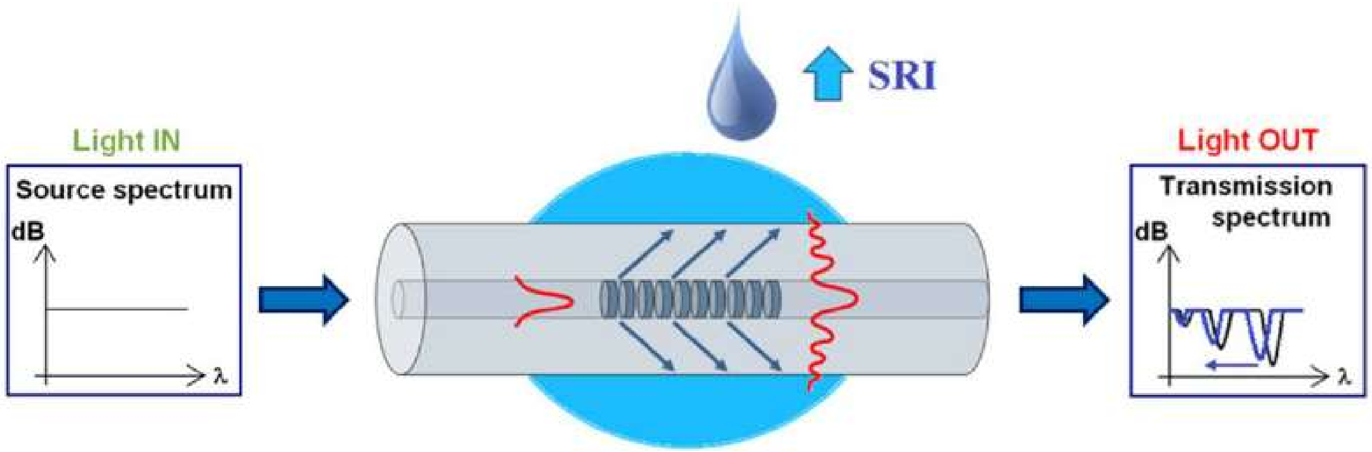
Method	RMS error (pm)	Testing Time (ms)
Long short-term memory	0.1165	8.82
Standard convolutional neural network	0.3467	14.63
Hybrid CNN-LSTM	0.809	148
Least squares	2.56	178,737
Dilated convolutional neural network	**0.0201**	**15.17**

Present work: Helical long-period grating

LPG for refractive index (RI) sensing

- Losses at cladding-air interface when light is guided from fundamental mode to cladding modes
- High attenuation for cladding modes: series of attenuation bands
- Transmission spectrum affected by changing of core/cladding guiding properties
- Strain, surrounding refractive index (SRI), temperature, torsion

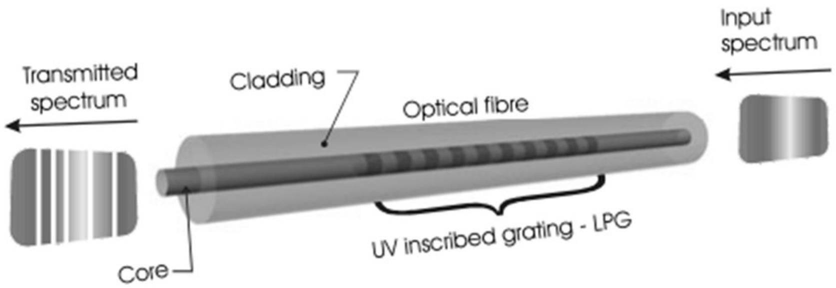
$$\lambda = \left[n_{eff}(\lambda) - n^{i}_{clad}(\lambda)\right]\Lambda$$

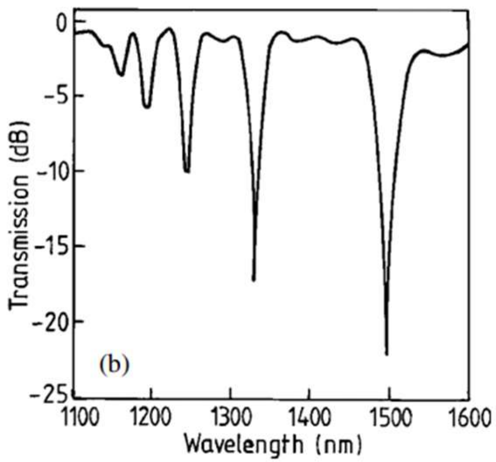


Present work: Helical long-period grating

Traditional fiber grating fabrication method: UV inscribed grating

- Sophisticated fabrication process
- Requires expensive phase mask
- UV laser is expensive

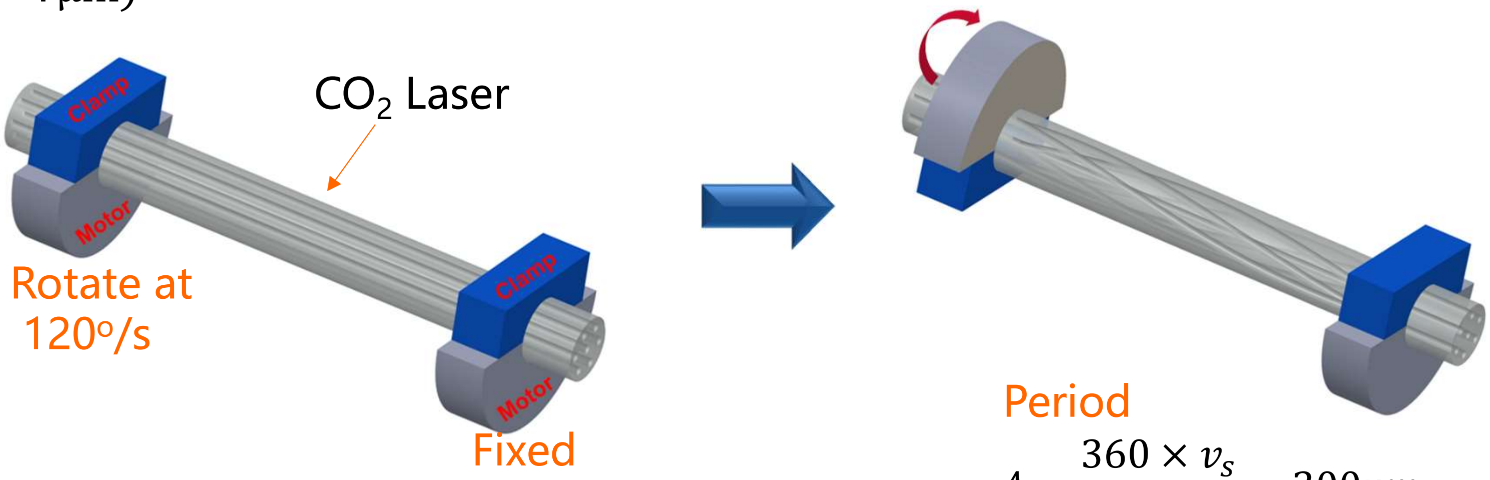


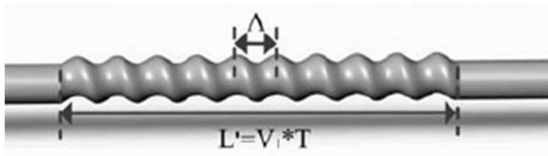


Present work: Helical long-period grating

To reduce the fabrication complexity and overall cost

- Novel way to produce fiber grating: Twisted fiber
- Limitation of machines: LPG could be achieved but not FBG (< 1μm)





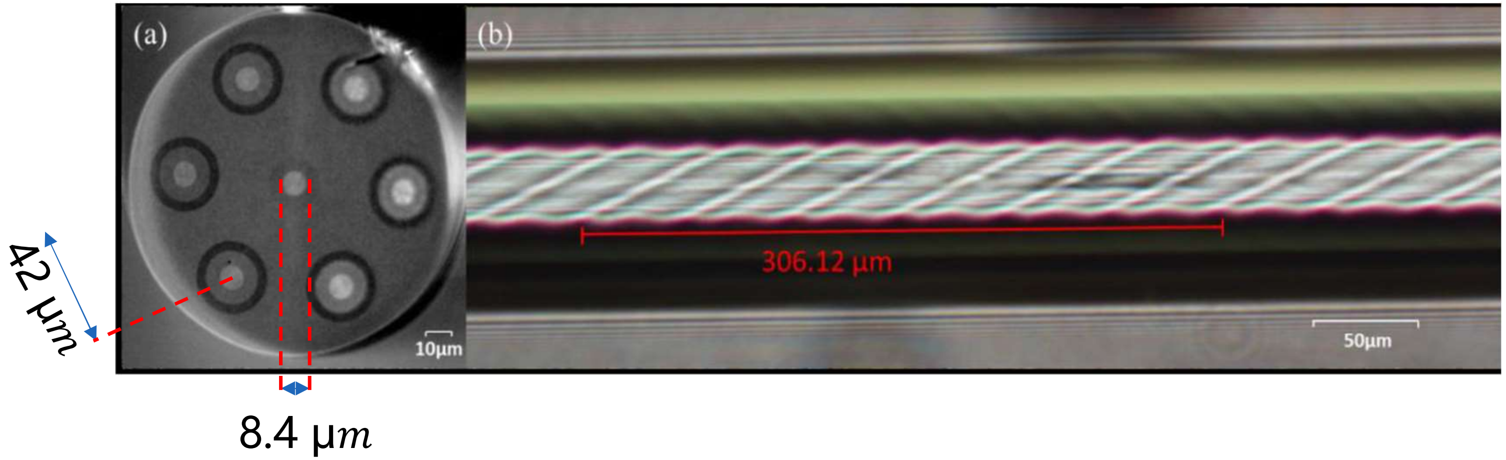
Period

$$\Lambda = \frac{360 \times v_s}{\omega_1 - \omega_2} = 300\mu m$$

Both motor shift towards right at v_s=0.1 $\mu m/ms$

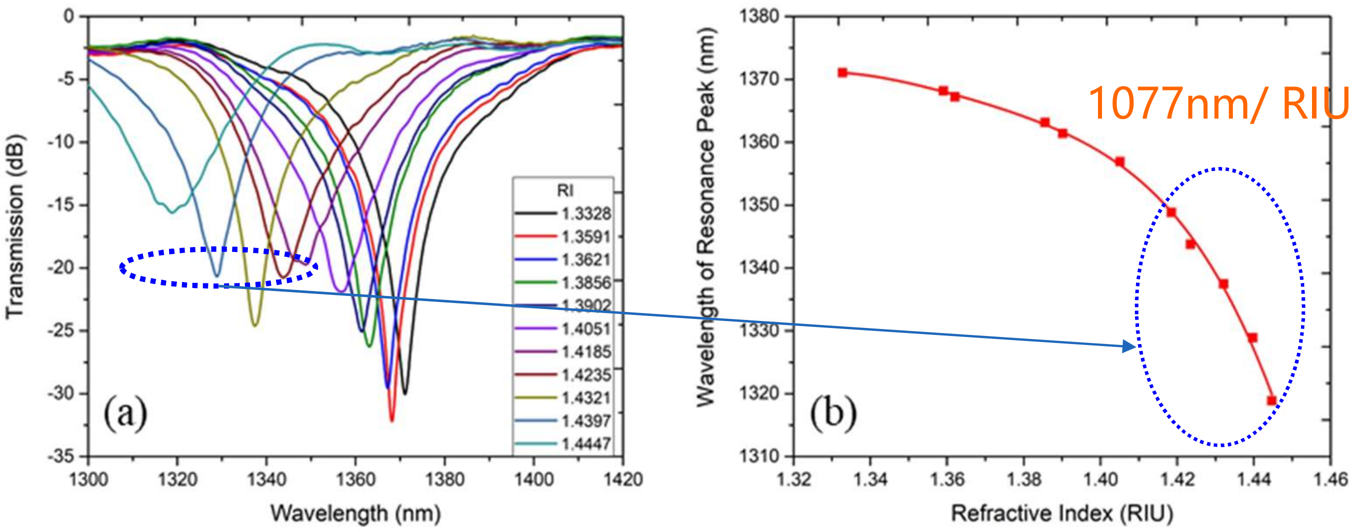
Present work: Helical long-period grating

The HLPG fabricated



Present work: Helical long-period grating

Experimental results

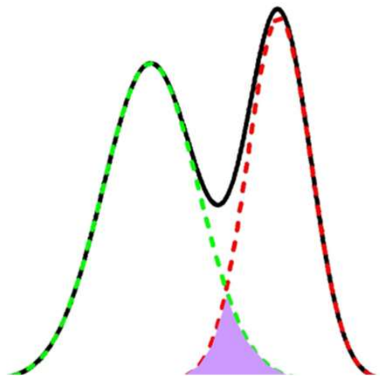


(a) Transmission spectra for serval refractive index values.

(b) Refractive index response

FBG overlapped signal demodulation

- We show that neural networks can distinguish highly overlapped signals
- Our dilated CNN model achieved record lowest RMS error of ~0.02 pm with competitive demodulation time of 15.17 ms (for 2 FBGs)
- Robust to noise
- Applicable to other peak detection problems



Helical long-period grating fiber

- Proposed fiber structure is easy to make, low-cost ar strong
- A high RI sensing of 1077nm/RIU was achieved
- Potential to increase the sensitivity with further process like etching, coating with metal film etc.